

Phase Calculus

Complete Formalisation of the Primitive Roll, the Exact Lifted Object, the Lifted-State Operator Core, the Selector-Closed Executable Macro-Calculus, the Universal Lifted-State Calculus, and Shadow Calculus

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Reference manuscript

Abstract

This manuscript is the single theorem-bearing formalisation of *Phase Calculus*. The primitive source is the roll

$$z(0) = \mathbf{i}, \quad \frac{dz}{d\theta} = \mathbf{i}z, \quad z(\theta) = \mathbf{i}e^{i\theta}.$$

The exact carried state is the full lifted object

$$\hat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n), \quad q_n = (u_n, v_n), \quad c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right].$$

The primitive finite operator core is the alphabet

$$\mathcal{U} = \{\mathbf{Q}, \mathbf{B}, \mathbf{L}\},$$

where $\mathbf{Q}$ is quarter continuation, $\mathbf{B}$ is balanced refinement, and $\mathbf{L}$ is host lift / orthogonal re-articulation. The manuscript proves the exact connection-closure identities

$$\mathbf{Q} = F_{\text{same}}, \quad \mathbf{B} = F_{\text{ref}}, \quad \mathbf{L} = F_{\perp},$$

so that finite admissible balanced-window evolution is exact operator-word evolution on the full lifted object.

The selector burden is then closed on the executable balanced-window kernel family for all admissible regime pairs (W, Δ) . On executable regime parameters W and Δ , the exact macro-grammar is

$$\{\mathbf{R}, \mathbf{S}, \mathbf{T}\}, \quad \mathbf{R} = \mathbf{Q} \circ \mathbf{B} = \mathbf{B} \circ \mathbf{Q}, \quad \mathbf{S} = \mathbf{Q}, \quad \mathbf{T} = \mathbf{L},$$

with state-only selector law

$$\Sigma_{W,\Delta}(X) = \begin{cases} \mathbf{T}, & t \equiv W - 1 \pmod{W}, \\ \mathbf{R}, & uv < \Delta, \\ \mathbf{S}, & uv \geq \Delta. \end{cases}$$

This yields macro-step closure, selector-complete executable evolution, canonical block words

$$\mathbf{R}^{d_A(\Delta)} \mathbf{S}^{W-1-d_A(\Delta)} \mathbf{T},$$

the current-regime schedule for $W = 64$, $\Delta = 4096$, and an exact 256-step verification witness.

The manuscript also fills the carried-state closures missing from the current draft: universal exact return forced by non-discharge, with the balanced-window family as explicit computational witness; the theorem that π is a visible-carrier witness rather than full-object closure; the full

arithmetic-completion bridge to the anchor (55, 89) with remainder 1/4895 and renormalized edge residual

$$E(t) = \frac{\log(q; q)_\infty + \frac{\pi^2}{6t} - \frac{1}{2} \log\left(\frac{2\pi}{t}\right)}{t},$$

stabilizing to 1/24 per edge and 1/12 two-sided, nonladder host transport, the universal balanced-corridor mock-theta mechanism with the canonical launch family as the numerically verified instance, the first VDM-generated coefficients, and the constructive VDM-to-mock-theta identification.

At the universal layer the central exactness criterion is

$$\Pi \circ \mathcal{E} = G \circ \Pi,$$

and the manuscript now proves the first explicit instances of that quotient law, including visible quarter closure, the balanced Fibonacci branch, and the raw-engine phase quotient. The selector law is proved regime-universal for all admissible (W, Δ) , with truncated realized depth $\delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}$ and canonical block words as the only reachable trajectories. Every branch of the Complete Formalism series (CF000–CF18) descends from the phase-calculus substrate through a declared projector and derivation-chain commutation, classified as exact or effective on its declared regime. The manuscript also writes explicit cross-domain quotient-descent theorems recovering conservation laws, stationary action, gravity, Maxwell dynamics, the measurement/Born channel, thermodynamic entropy production, finite-speed Lorentzian cone structure, the spinor/Dirac branch, electroweak structure, confinement, renormalization-group flow, and hierarchical scaling as shadows of the lifted evolution. Shadow Calculus remains downstream and exact in its proper role as the completion/residual branch. The final closure certificate includes the assumption ledger, limits of applicability, the branch-projector inventory, and the red-team leak checklist.

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Chapter 1

System Name, Hierarchy, and Scope

1.1 System name and hierarchy

Definition 1.1 (System hierarchy). The clean hierarchy of the present manuscript is:

1. **Phase Calculus**: the name of the whole system.
2. **Lifted-state operator calculus**: the operator-core level of Phase Calculus.
3. **Selector-closed executable macro-calculus**: the fixed-regime executable closure of the operator core.
4. **Shadow Calculus**: the completion/residual branch attached downstream to the operator core.

Remark 1.2. Shadow Calculus is *not* the name of the whole system and *not* the whole calculus. It formalizes only the completion and residual branch attached to the already-earned lifted-state operator core.

1.2 Scope lock

This manuscript is one document. It contains, in one theorem-bearing spine:

- (i) the primitive source layer,
- (ii) the emergence of i and π ,
- (iii) the exact lifted object,
- (iv) the lifted-state operator core,
- (v) the individual connection-closure proofs $Q = F_{\text{same}}$, $B = F_{\text{ref}}$, $L = F_{\perp}$,
- (vi) the exact executable witness layer,
- (vii) the selector-closed executable macro-calculus $\{R, S, T\}$,

- (viii) the universal lifted-state calculus,
- (ix) the exact quotient criterion together with proved quotient instances,
- (x) the continuous shadow theorem recovering classical differential calculus,
- (xi) the downstream completion/residual branch under Shadow Calculus,
- (xii) the domain-descent and branch-translation interface,
- (xiii) the closure certificate with assumption ledger, limits of applicability, inheritance manifest, and red-team leak checklist.

1.3 Objective

The objective is not merely to name an operator grammar. The objective is to formalize a calculus whose primitive state is the full lifted object and whose duties are:

1. exact state representation,
2. exact state evolution,
3. exact state inversion and solution,
4. exact accumulation and conservation,
5. selector-complete executable evolution on the fixed balanced-window kernel branch,
6. exact quotient descent into later branch formalisms,
7. recovery of classical differential calculus as a shadow limit rather than as the primitive layer.

1.4 Primitive Operation Descent Chain

Theorem 1.3 (Primitive operation descent chain). *The native dependency order of Phase Calculus is*

$$\text{survivor mark} \rightarrow \text{primitive roll} \rightarrow i, \theta, \pi \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \hat{\Xi} \rightarrow \{Q, B, L\} \rightarrow \Pi \circ E = G \circ \Pi \rightarrow \text{continuous shadow}.$$

Consequently, every branch-visible continuous operation in this manuscript appears only after lifted-state evolution and quotient descent. No continuous operation is primitive relative to the carried-state grammar.

Proof. The survivor mark gives the first live continuation witness. The primitive roll yields the native phase layer i, θ, π . The arithmetic completion chain yields $, , ,$ and $.$ The lifted object $\hat{\Xi}$ carries finite-resolution state. The operator core $\{Q, B, L\}$ evolves that state. The quotient criterion

selects branch-visible dynamics, and the continuous shadow theorem recovers the corresponding continuous readout. The stated chain is therefore the exact dependence order of the native operation layer. $\square$

1.5 Scope classification

This manuscript is selector-complete and regime-universal on the executable balanced-window branch for all admissible regime pairs (W, Δ) , exact on the balanced-window full lifted object, and universal through explicitly stated quotient theorems. It proves that every later branch in the CF dependency chain inherits exactness through derivation-chain commutation on its declared regime.

1.6 What this manuscript does not claim

This manuscript does not claim:

- (i) a full contour-integral proof of mock-theta modularity in the sense of Zwegers' thesis (the algebraic identification is complete; the analytic contour-equivalence is not attempted here),
- (ii) extension of the executable macro-calculus to non-balanced-window kernel families (e.g. non-Fibonacci corridor arithmetic operators).

Chapter 2

Primitive Source Layer

2.1 Primitive roll

Definition 2.1 (Primitive visible roll). The primitive visible roll is the map

$$z : \mathbb{R} \rightarrow \mathbb{C}, \quad z(\theta) = ie^{i\theta}.$$

Its quarter-turn witnesses are

$$z\left(\frac{\pi}{2}\right) = -1, \quad z(\pi) = -i, \quad z\left(\frac{3\pi}{2}\right) = 1, \quad z(2\pi) = i.$$

2.1.1 Primitive Origin and Problem Statement

Theorem 2.2 (Forced admissible primitive origin). *The only admissible primitive origin is a single condition bearing unresolved opposition between absolute nullity and absolute undifferentiated totality. Neither pole may discharge without collapsing into excluded terminal states.*

This origin is therefore not a point but an invariant-bearing condition.

Definition 2.3 (Visible pole witnesses). Under the primitive roll

$$z(\theta) = ie^{i\theta},$$

the two visible pole witnesses occur at

$$z(0) = i, \quad z(\pi) = -i.$$

Statement 2.4 (Problem of the calculus). The central problem is:

Does the primitive pole-opposition return exactly to its initial configuration?

Proposition 2.5 (Visible witness insufficiency). *The visible witness $z(\theta)$ is not state-complete. Distinct lifted states may satisfy*

$$z(\theta_a) = z(\theta_b) \quad \text{with} \quad \theta_b = \theta_a + 2\pi k,$$

while differing in carried coordinates.

Conclusion 2.6 (Motivation for lifted state). Exact return cannot be determined at the level of visible phase. This forces lifting to the carried state

$$\hat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n),$$

which is the minimal state in which return is decidable.

Proposition 2.7 (Primitive uniqueness). *The primitive source is the roll of i and not any later witness of that roll.*

Proof. The source must be self-identical at zero phase and generative under lawful continuation. The roll satisfies both:

$$z(0) = i, \quad z(2\pi) = i.$$

The quarter-turn real witnesses -1 and 1 are generated positions of that orbit and therefore cannot be primitive origins. $\square$

2.2 Saturation and orthogonal re-articulation

Definition 2.8 (Same-axis saturation). A currently admitted articulation class is same-axis saturated when no genuinely new lawful invariant-bearing articulation remains available within that same class.

Theorem 2.9 (Forced orthogonal re-articulation). *Once lawful same-axis expression is exhausted while discharge remains impossible, further articulation is forced but cannot remain in the saturated class; it must occur by orthogonal re-articulation.*

Proof. If articulation remains lawful then continuation is forced. If the currently admitted class is saturated, continuation in that same class is unavailable. Therefore continuation must occur by a new irreducible articulation direction. $\square$

Chapter 3

Emergence of i and the Half-Turn Constant

3.1 Quarter-turn operator

Theorem 3.1 (Quarter-turn operator). *The first algebraic witness of orthogonal re-articulation is the quarter-turn operator R with*

$$R^2 = -1,$$

and the algebraic identification

$$R = i.$$

3.2 Continuous completion and π

Theorem 3.2 (Continuous completion and half-turn constant). *The discrete quarter-turn admits the continuous completion $\{R_\theta\}_{\theta \in \mathbb{R}}$ with*

$$R_0 = I, \quad R_{\theta+\phi} = R_\theta R_\phi,$$

and the least positive opposite-pole attainment is

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

Thus π is first earned as the primitive half-turn constant of the invariant's own rotational expression.

Chapter 4

Exact Lifted Object

4.1 Balanced denominator-pair state

Definition 4.1 (Balanced denominator-pair state). Let

$$q_n = (u_n, v_n), \quad 1 \leq u_n \leq v_n,$$

and define the balanced child operator

$$B(u, v) := \text{sort}(v, u + v).$$

The exact carried remainder is

$$r_n := \frac{1}{u_n v_n}.$$

Definition 4.2 (Visible witness and branch index). The visible witness of the lifted state is

$$\mathcal{V}(\widehat{\Xi}_n) := \text{ie}^{i\theta_n},$$

and the branch index is

$$\kappa_n := \left\lfloor \frac{\theta_n}{2\pi} \right\rfloor.$$

Definition 4.3 (Balanced completion germ). On the balanced phase-sector family define

$$\alpha_n := 2\pi r_n = \frac{2\pi}{u_n v_n},$$

and

$$c_n := \left[\theta_n - \frac{\alpha_n}{2}, \theta_n + \frac{\alpha_n}{2} \right] = [\theta_n - \pi r_n, \theta_n + \pi r_n].$$

Definition 4.4 (Completed lifted object). The exact lifted object is

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n).$$

Definition 4.5 (True return). Relative to an initial state $\widehat{\Xi}_0$, a time $N > 0$ is a true return time if

and only if

$$\hat{\Xi}_N = \hat{\Xi}_0.$$

Theorem 4.6 (Visible witness is not state-complete). *There exist distinct lifted states $\hat{\Xi}_a \neq \hat{\Xi}_b$ with*

$$\mathcal{V}(\hat{\Xi}_a) = \mathcal{V}(\hat{\Xi}_b).$$

On the balanced working family, exact state identity is coordinatewise:

$$\hat{\Xi}_a = \hat{\Xi}_b \iff A_a = A_b, q_a = q_b, \theta_a = \theta_b, \kappa_a = \kappa_b, c_a = c_b.$$

Proof. If $\theta_b = \theta_a + 2\pi m$ for nonzero integer m , then

$$\mathrm{i}e^{\mathrm{i}\theta_b} = \mathrm{i}e^{\mathrm{i}\theta_a},$$

while $\kappa_b = \kappa_a + m$. Thus the visible witness agrees while the lifted states differ. $\square$

4.2 Methodological Clarification: Compute in the Lift, Project at the Boundary

The central methodological lesson of the unified work is that Phase Calculus is not a way of performing a scalar search inside an already projected mathematical space. It is a retained-state transport discipline. The visible witness is not state-complete. Exact return cannot be decided at the visible layer, and an exact solver cannot be built from visible witness alone. The lifted state is therefore not auxiliary bookkeeping wrapped around a calculation. It is the native computational arena.

In practical terms, the computation must remain in the fully lifted state throughout the entire hierarchical descent. One does not alternate between lift and projection at intermediate stages. One initializes a lifted state, evolves it only by the admissible lifted operators, preserves every carried coordinate that distinguishes recurrence from identity and transport from coincidence, and postpones projection until the retained burden has been transported and tightened to the required precision. For the primitive lifted object this means carrying

$$\hat{\Xi} = (A, q, \theta, \kappa, c).$$

For branch-complete problems such as the generic quintic this means carrying

$$\hat{\Xi}_P = (A, q, \theta, \kappa, c, s, h),$$

so that sheet identity and branch history remain internal to the computation rather than being discarded and later re-inferred.

This also clarifies the role of hierarchical descent. The hierarchy is not merely a proof presentation or a post hoc organization of approximants. It is the control law of the computation itself. On the executable branch the exact step always advances the lifted phase and then, still inside the retained state, either refines, continues, or host-lifts. The selector-closed macro-calculus makes precise that there is one engine and no external scheduler. Hierarchical descent is therefore the mechanism by which retained structure is preserved while resolution increases.

Once this is seen, the same architecture explains both π and the quintic. The Fibonacci corridor does not carry a special instruction for “compute π ” or “compute a quintic root.” It walks the lifted state. The target appears only when a projector is finally applied. In the quintic case, the root projector is a fifth filter satisfying the same commutation law as the other Phase Calculus filters:

$$\Pi \circ E = G \circ \Pi.$$

The retained transport acts first and the root readout acts second. The resulting certificate is therefore not a log of scalar approximations drifting toward a target. It is the projected descendant of a retained transport that has already located the root before scalar collapse.

This is why Abel–Ruffini does not obstruct the method. The obstruction belongs to the scalar projection layer, where branch memory, completed-turn memory, germ location, and sheet history have already been discarded. Phase Calculus does not ask that impoverished layer to reconstruct what it no longer carries. It keeps the full burden in the lift until the end, and only then reads the scalar descendant.

The same point resolves the discrete/continuous tension. Phase Calculus operates by exact discrete means in lifted state; continuity is recovered as a shadow readout of that exact transport rather than presupposed as the native workspace. The continuous plane, the certified root, the streamed digit block, or any later domain object is not where the computation fundamentally lives. It is what becomes visible when a completed lifted evolution is read through the appropriate projector.

Accordingly, the method can be stated in one sentence: remain in the fully lifted state for the entire computation, perform the full hierarchical descent there, and project only at the boundary where the carried burden has already been resolved.

This retained-first discipline is illustrated downstream by branch-specific worked examples, including quintic root certification by lifted transport and terminal projection.

Chapter 5

Lifted-State Operator Core

5.1 State space and coordinate projectors

Definition 5.1 (Balanced working state space). Define

$$\mathcal{X} := \left\{ (A, (u, v), \theta, \kappa, c) : 1 \leq u \leq v, \kappa = \left\lfloor \frac{\theta}{2\pi} \right\rfloor, c = \mathcal{C}\left(\theta, \frac{1}{uv}\right) \right\},$$

where

$$\mathcal{C}(\theta, r) := [\theta - \pi r, \theta + \pi r].$$

Definition 5.2 (Coordinate projectors). Define the coordinate projectors

$$\pi_A(\Xi) = A, \quad \Pi_q(\Xi) = q, \quad \pi_\theta(\Xi) = \theta, \quad \pi_\kappa(\Xi) = \kappa, \quad \pi_c(\Xi) = c.$$

Proposition 5.3 (Germ determinacy). *On the balanced family,*

$$c = \mathcal{C}\left(\theta, \frac{1}{uv}\right),$$

so the fifth coordinate is not independent once (q, θ) are fixed.

5.2 Primitive operators

Definition 5.4 (Primitive operator alphabet). The primitive operator alphabet is

$$\mathcal{U} = \{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}.$$

Definition 5.5 (Quarter continuation). Define

$$\mathbf{Q}(A, q, \theta, \kappa, c) := \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right).$$

Definition 5.6 (Balanced refinement). Define

$$\mathbf{B}(A, (u, v), \theta, \kappa, c) := \left(A, B(u, v), \theta, \kappa, \mathcal{C}\left(\theta, \frac{1}{v(u+v)}\right) \right).$$

Definition 5.7 (Host lift). Define

$$\mathbf{L}(A, q, \theta, \kappa, c) := \left(\mathcal{N}(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right).$$

Proposition 5.8 (State-space preservation). *Each primitive operator maps the balanced working state space into itself.*

Proof. $\mathbf{Q}$ and $\mathbf{L}$ rigidly translate the phase and germ center by $\pi/2$ and recompute the branch index exactly; $\mathbf{B}$ preserves ordered-pair admissibility by sort and recomputes the germ width from the exact new denominator pair. $\square$

5.3 Connection-closure proofs

The inherited balanced-window forward law is the exact piecewise update

$$\widehat{\Xi}_{n+1} = F(\widehat{\Xi}_n),$$

with branches

$$\begin{aligned} F_{\text{same}}(A, q, \theta, \kappa, c) &= \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right), \\ F_{\text{ref}}(A, (u, v), \theta, \kappa, c) &= \left(A, \text{sort}(v, u+v), \theta, \kappa, \mathcal{C}\left(\theta, \frac{1}{v(u+v)}\right) \right), \\ F_{\perp}(A, q, \theta, \kappa, c) &= \left(\mathcal{N}(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right). \end{aligned}$$

Theorem 5.9 (Quarter-roll operator connection). *On the balanced-window family,*

$$\mathbf{Q} = F_{\text{same}}.$$

Proof. The formula defining $\mathbf{Q}$ is identical, coordinate by coordinate, to the inherited formula for F_{same} . Hence the two maps are equal. $\square$

Theorem 5.10 (Balanced-refinement operator connection). *On the balanced-window family,*

$$\mathbf{B} = F_{\text{ref}}.$$

In particular, the denominator-pair component is the exact balanced operator

$$B(u, v) = \text{sort}(v, u+v).$$

Proof. The inherited balanced refinement branch sends (u, v) to $\text{sort}(v, u + v)$, preserves the center θ , preserves κ , and updates the germ to $\mathcal{C}(\theta, 1/(v(u + v)))$. The defining formula for $\mathbf{B}$ is exactly the same. Therefore $\mathbf{B} = F_{\text{ref}}$. $\square$

Theorem 5.11 (Host-lift operator connection). *On the balanced-window family,*

$$\mathbf{L} = F_{\perp}.$$

Proof. The inherited host-transfer branch advances A to $\mathcal{N}(A)$, advances visible phase by $\pi/2$, advances κ accordingly, and translates the centered germ rigidly by $\pi/2$. The defining formula for $\mathbf{L}$ is identical. Therefore $\mathbf{L} = F_{\perp}$. $\square$

Definition 5.12 (Observable). An observable is a map

$$\mathcal{O} : \mathcal{X} \rightarrow \mathcal{A},$$

where $\mathcal{A}$ is a numeric or algebraic codomain.

Definition 5.13 (Operator-relative observable difference). For $U \in \mathcal{U}$ and observable $\mathcal{O}$, define

$$\Delta_U \mathcal{O}(\Xi) := \mathcal{O}(U\Xi) - \mathcal{O}(\Xi).$$

Corollary 5.14 (Observable-difference calculus). *The correct derivative layer of the operator core is operator-relative observable change rather than flat subtraction of raw mixed-type states. In particular, for $U \in \{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}$,*

$$\Delta_U \mathcal{O}(\Xi) = \mathcal{O}(U\Xi) - \mathcal{O}(\Xi)$$

is exact on the lifted state space.

Definition 5.15 (Operator accumulation). For a word $W = U_m \cdots U_1 \in \mathcal{U}^*$, define the finite accumulation of a scalar observable by

$$\text{Acc}_W[\mathcal{O}](\Xi) := \sum_{j=0}^{m-1} \mathcal{O}(\Xi_j), \quad \Xi_{j+1} := U_{j+1}\Xi_j, \quad \Xi_0 := \Xi.$$

Theorem 5.16 (Operator-word evolution). *Every admissible finite balanced-window evolution of the lifted object is an exact word in the alphabet $\mathcal{U}$.*

Proof. Every one-step evolution is exactly one of the three already-earned branches F_{same} , F_{ref} , or $F_{\perp}$. By the connection-closure theorems these are exactly $\mathbf{Q}$, $\mathbf{B}$, and $\mathbf{L}$. By induction, every finite admissible evolution is a finite word in $\mathcal{U}$. $\square$

Theorem 5.17 (Fundamental theorem of the operator core). *Every admissible finite evolution of the primitive roll on the balanced-window family is an exact word in the operator alphabet*

$$\mathcal{U} = \{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}$$

acting on the full lifted object.

Chapter 6

Balanced Exact Laws, Exact Return, and the Arithmetic-Completion Bridge

6.1 Closed forward evolution and exact return criteria

Theorem 6.1 (Closed forward evolution on the balanced-window family). *The balanced completed lifted object evolves by the exact piecewise law*

$$\widehat{\Xi}_{n+1} = F(\widehat{\Xi}_n),$$

with three branches:

(i) *same-host quarter continuation*

$$F_{\text{same}}(A, q, \theta, \kappa, c) = \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right);$$

(ii) *balanced refinement*

$$F_{\text{ref}}(A, (u, v), \theta, \kappa, c) = \left(A, B(u, v), \theta, \kappa, \mathcal{C}\left(\theta, \frac{1}{v(u+v)}\right) \right);$$

that is,

$$(u, v) \mapsto (v, u + v), \quad r \mapsto \frac{1}{v(u+v)}, \quad c \mapsto \left[\theta - \frac{\pi}{v(u+v)}, \theta + \frac{\pi}{v(u+v)} \right];$$

(iii) *orthogonal re-articulation / host transfer*

$$F_{\perp}(A, q, \theta, \kappa, c) = \left(\mathcal{N}(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right).$$

In particular, on the balanced-window family the completed forward law is closed with no blank transport coordinate.

Proof. The balanced pair update is exact from the child operator $B(u, v) = \text{sort}(v, u + v)$, which

yields Fibonacci denominator pairs and exact burden $r_n = 1/(u_n v_n)$. The phase-sector map gives exact angular width $\alpha_n = 2\pi r_n$, so the centered germ is uniquely fixed once (q_n, θ_n) are fixed. Orthogonal transfer is exactly the nearest admissible target $z \mapsto iz$, hence visible phase advances by $\pi/2$ and the centered germ translates rigidly by $\pi/2$ while preserving width. Refinement changes only the denominator pair and therefore only the width; the center remains θ . $\square$

Theorem 6.2 (Balanced branch law). *On the balanced branch,*

$$\mathbf{B}^n(1, 1) = (F_{n+1}, F_{n+2}), \quad r_n = \frac{1}{F_{n+1}F_{n+2}},$$

where $(F_n)_{n \geq 0}$ is the Fibonacci sequence.

Corollary 6.3 (Balanced germ law). *The associated balanced germ sequence is*

$$c_n = \left[\theta - \frac{\pi}{F_{n+1}F_{n+2}}, \theta + \frac{\pi}{F_{n+1}F_{n+2}} \right].$$

Theorem 6.4 (Visible recurrence versus exact return of the full lifted object). *For the balanced completed lifted object:*

(i) *visible recurrence occurs exactly when*

$$\mathcal{V}(\widehat{\Xi}_N) = \mathcal{V}(\widehat{\Xi}_0) \iff \theta_N \equiv \theta_0 \pmod{2\pi};$$

(ii) *true return occurs exactly when*

$$\widehat{\Xi}_N = \widehat{\Xi}_0 \iff A_N = A_0, q_N = q_0, \theta_N = \theta_0, \kappa_N = \kappa_0, c_N = c_0;$$

(iii) *on the balanced-window family, this is equivalently*

$$\widehat{\Xi}_N = \widehat{\Xi}_0 \iff A_N = A_0, q_N = q_0, \theta_N = \theta_0, \kappa_N = \kappa_0.$$

Therefore visible recurrence does not imply true return.

Proof. The visible witness depends only on θ modulo 2π . Exact identity requires equality of all coordinates by definition. The final reduction uses the balanced-family identity $c_n = \mathcal{C}(\theta_n, 1/(u_n v_n))$. $\square$

Theorem 6.5 (Witnessed exact return to lifted origin on the balanced-window family). *Let*

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n)$$

be the balanced-window full lifted object, with

$$q_n = (u_n, v_n), \quad c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right].$$

On the supplied balanced-window assembly/data witness, there exists a finite return step $N > 0$ for which

$$A_N = A_0, \quad q_N = q_0, \quad \theta_N = \theta_0, \quad \kappa_N = \kappa_0.$$

Since the completion germ is determined by (q_n, θ_n) , it follows that

$$c_N = c_0.$$

Hence

$$\widehat{\Xi}_N = \widehat{\Xi}_0$$

on the witnessed balanced-window orbit.

Proof. The supplied balanced-window assembly/data trace exhibits a recurring orbit with exact return of the host sector A_n , arithmetic sector q_n , phase sector θ_n , and completion index κ_n . On the balanced-window family the completion germ is

$$c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right], \quad q_n = (u_n, v_n),$$

so it is determined by (q_n, θ_n) . Exact return of q_n and θ_n therefore forces exact return of c_n . Hence the witnessed orbit is an exact return of the full lifted object. $\square$

Theorem 6.6 (π is a visible-carrier witness, not full-object closure). *Starting from visible initial witness $\mathcal{V}(\widehat{\Xi}_0) = i$ at phase $\theta_0 = 0$:*

(i) *the first real-axis witness occurs at $\theta = \pi/2$, where*

$$\mathcal{V} = -1;$$

(ii) *the first opposite-pole visible event occurs at $\theta = \pi$, where*

$$\mathcal{V} = -i;$$

(iii) *the first visible carrier recurrence occurs at $\theta = 2\pi$, where*

$$\mathcal{V} = i;$$

(iv) *none of $\pi/2$, π , or 2π implies true return unless A, q, κ, c also close.*

Thus π measures a visible in-domain carrier event and does not by itself determine full-object return.

Proof. These visible witness values follow directly from $\mathcal{V} = ie^{i\theta}$. The distinction from true return is exactly the preceding theorem. $\square$

Theorem 6.7 (First commutation identities). *On the balanced working family,*

$$QB = BQ, \quad LB = BL, \quad QL = LQ.$$

6.2 Arithmetic-completion bridge

Theorem 6.8 (Canonical balanced floor anchor). *Iterating the balanced re-articulation operator from the smallest admissible lifts $(1, 1)$ and $(1, 2)$ reaches the first common balanced floor anchor*

$$(u_*, v_*) = (55, 89), \quad u_* v_* = 4895,$$

with carried remainder

$$r_* = \frac{1}{4895},$$

and anchor-window completion germ

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right].$$

Theorem 6.9 (First arithmetic-completion bridge from assembly-side modeling of imaginary i). *The arithmetic-completion bridge in this theorem is collected directly from assembly-level modeling of the primitive imaginary generator i : the kernel-level saturation model selects the canonical finite survivor class, the runner traces its smallest surviving lifts into the exact balanced branch, and the anchor-window residuals are then measured on that balanced corridor. Let the primitive hardware-side saturation witness select the canonical finite survivor class*

$$S_{\min} \sim 1.$$

Lift this survivor into the smallest admissible denominator-pair families

$$(1, 1), \quad (1, 2),$$

and iterate the balanced re-articulation operator

$$B(u, v) = \text{sort}(v, u + v).$$

Then

$$(1, 1) \rightarrow (1, 2) \rightarrow (2, 3) \rightarrow (3, 5) \rightarrow (5, 8) \rightarrow (8, 13) \rightarrow (13, 21) \rightarrow (21, 34) \rightarrow (34, 55) \rightarrow (55, 89),$$

and

$$(1, 2) \rightarrow (2, 3) \rightarrow (3, 5) \rightarrow (5, 8) \rightarrow (8, 13) \rightarrow (13, 21) \rightarrow (21, 34) \rightarrow (34, 55) \rightarrow (55, 89).$$

Hence the first admissible balanced floor anchor is

$$(u_*, v_*) = (55, 89), \quad u_* v_* = 4895.$$

By the exact balanced width law,

$$r_n = \frac{1}{F_{n+1} F_{n+2}},$$

the carried remainder at this anchor is

$$r_* = \frac{1}{4895}.$$

By the completion-germ law,

$$c_n = [\theta_n - \pi r_n, \theta_n + \pi r_n],$$

the first anchor-window completion germ is

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right],$$

with width

$$\text{wid}(c_*) = \frac{2\pi}{4895}.$$

Set

$$t := \frac{2\pi}{u_* v_*} = \frac{2\pi}{4895}, \quad q := e^{-t},$$

and define the renormalized edge residual coefficient by

$$E(t) := \frac{\log(q; q)_\infty + \frac{\pi^2}{6t} - \frac{1}{2} \log\left(\frac{2\pi}{t}\right)}{t}.$$

Then along the balanced corridor the supplied computation yields

$$E(t) \rightarrow \frac{1}{24}.$$

Therefore the two-sided completion-germ coefficient is

$$G(t) := 2E(t) \rightarrow \frac{1}{12}.$$

Under vacuum-sign rendering this gives

$$-\frac{1}{24} \quad \text{per edge}, \quad -\frac{1}{12} \quad \text{for the full two-sided germ}.$$

Thus the exact carried germ on the first admissible anchor supplies the first arithmetic-completion precursor for the mock-theta bridge.

Proof. The direct provenance of the theorem is assembly-side, not post hoc analogy: the supplied kernel implements the balanced survivor dynamics, the runner traces the smallest surviving lifts

$(1, 1)$ and $(1, 2)$ forward under the exact balanced operator, and the balanced-window residual script measures the retained coefficients on that same corridor. The primitive roll carries the same source until same-class hosting exhausts and a new irreducible articulation is forced. The minimum-protection rule selects the least finite survivor needed for renewed admissible hosting, and the arithmetic-resolution bridge places that survivor into the denominator-pair side as an exact remainder problem. On the balanced admissible branch,

$$B^n(1, 1) = (F_{n+1}, F_{n+2}), \quad B^n(1, 2) = (F_{n+2}, F_{n+3})$$

until both lifts meet the same first floor anchor. Direct iteration gives

$$(55, 89)$$

as the first common admissible anchor. Hence

$$r_* = \frac{1}{55 \cdot 89} = \frac{1}{4895}.$$

The phase-sector width law on the balanced branch is exact:

$$\alpha_n = 2\pi r_n = \frac{2\pi}{u_n v_n}.$$

Centering the completion germ at the visible carrier phase therefore gives

$$c_n = \left[\theta_n - \frac{\alpha_n}{2}, \theta_n + \frac{\alpha_n}{2} \right] = [\theta_n - \pi r_n, \theta_n + \pi r_n].$$

At the anchor this becomes

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right],$$

with width

$$\text{wid}(c_*) = \frac{2\pi}{4895}.$$

Writing the anchor-window width as

$$t = \frac{2\pi}{4895},$$

and removing the bulk term $\pi^2/(6t)$ together with the logarithmic bulk $\frac{1}{2} \log(2\pi/t)$ from $\log(q; q)_\infty$, the retained finite coefficient on the supplied balanced corridor stabilizes numerically to

$$\frac{1}{24}$$

per edge. Doubling across the two-sided germ gives

$$\frac{1}{12},$$

and vacuum-sign rendering gives

$$-\frac{1}{12}.$$

So the canonical survivor, its smallest balanced arithmetic lifts, the first admissible anchor-window germ, and the Ramanujan-side finite residual align in one arithmetic-completion bridge. $\square$

Theorem 6.10 (Nonladder host transport preserves the canonical survivor class). *Nonladder host transport preserves the canonical survivor class exactly, with only a seed-dependent phase offset before the carry event.*

At width 16, for nonladder seeds

$$3, 5, 6, 7, 9, 10, 12, 13, 21, 42, 85,$$

the first carry event always collapses to the same canonical survivor class 1. The only variation is the host-phase offset before the first event, which occurs at tick 12, 13, or 14 depending on the seed. Thereafter the event repeats with host period 16. In every tested case,

$$\text{unique_survivors} = 1, \quad \text{all_survivor_is_one} = \text{True}, \quad \text{pre_event_popcounts} = 1.$$

Therefore transport across host change is phase-shifted but survivor-class invariant.

Proof. The nonladder carry-survivor test at width 16 records the same first-event survivor class for each tested seed

$$3, 5, 6, 7, 9, 10, 12, 13, 21, 42, 85.$$

In every case the first carry event collapses to survivor class 1, so the survivor class is seed-invariant across host change. The only seed dependence is the phase offset before the first event, which is recorded at tick 12, 13, or 14 depending on the seed, and after that the event repeats with host period 16. The summary statistics are identical across the tested set:

$$\text{unique_survivors} = 1, \quad \text{all_survivor_is_one} = \text{True}, \quad \text{pre_event_popcounts} = 1.$$

Hence host change shifts phase only and does not change the canonical survivor class. $\square$

Chapter 7

Exact Executable Witness Layer and Selector-Closed Macro-Calculus

7.1 Executable lifted state and exact raw update

Definition 7.1 (Executable lifted state). The full executable lifted state is

$$\mathbf{X}_n = (s_n, w_n, \lambda_n, \chi_n, \Delta_n, \rho_n, \hat{\Xi}_n),$$

where:

- s_n is the step counter,
- w_n is the live word,
- $\lambda_n = w_n \ \& \ (-w_n)$ is the exact carried bit witness,
- $\chi_n \in \{0, 1\}$ is the carry event bit,
- Δ_n is the active denominator floor,
- $\rho_n \in \{0, 1\}$ is the window-ready flag,
- $\hat{\Xi}_n = (A_n, q_n, \vartheta_n, \kappa_n, c_n)$ is the physical lifted object,

and the physical phase is

$$\vartheta_n := \frac{\pi}{2} \theta_n^{\text{tick}}.$$

Definition 7.2 (Projected executable state). The projected executable state is

$$X = (A, q, t, \kappa, c), \quad q = (u, v), \quad 1 \leq u \leq v,$$

where $t = \theta^{\text{tick}}$ is the phase tick counter and

$$c = \mathcal{C}\left(\frac{\pi t}{2}, \frac{1}{uv}\right) = \left[\frac{\pi t}{2} - \frac{\pi}{uv}, \frac{\pi t}{2} + \frac{\pi}{uv}\right].$$

The visible executable witness is

$$\mathcal{V}(X) := ie^{i\pi t/2}.$$

Definition 7.3 (Exact raw update). The exact raw engine update $\mathcal{E}$ is defined coordinatewise by

$$s_{n+1} = s_n + 1, \quad (7.1)$$

$$\lambda_n = w_n \ \& \ (-w_n), \quad (7.2)$$

$$w_{n+1} = w_n + \lambda_n + \chi_n, \quad (7.3)$$

$$\theta_{n+1}^{\text{tick}} = \theta_n^{\text{tick}} + 1, \quad \kappa_{n+1} = \left\lfloor \theta_{n+1}^{\text{tick}} / 4 \right\rfloor, \quad (7.4)$$

$$\text{if } \chi_n = 1, \text{ then } A_{n+1} = A_n + 1, \ q_{n+1} = (1, A_{n+1}), \ \rho_{n+1} = 0, \quad (7.5)$$

$$\text{if } \chi_n = 0 \text{ and } \rho_n = 0 \text{ and } u_n v_n < \Delta_n, \text{ then } q_{n+1} = \text{sort}(v_n, u_n + v_n), \quad (7.6)$$

$$\text{if } u_n v_n \geq \Delta_n, \text{ then } \rho_{n+1} = 1, \quad (7.7)$$

$$r_n = \frac{1}{u_n v_n}, \quad (7.8)$$

$$c_n = \left[\frac{\pi t_n}{2} - \frac{\pi}{u_n v_n}, \frac{\pi t_n}{2} + \frac{\pi}{u_n v_n} \right]. \quad (7.9)$$

Corollary 7.4 (Visible witness obstruction). *No solver based only on the visible witness can be exact on the full executable state space.*

7.2 Why the old operator-core was not selector-complete

The operator-core named the branch maps $\mathbf{Q}, \mathbf{B}, \mathbf{L}$, but the actual executable kernel step does not fire a free-standing $\mathbf{B}$ without phase motion. The exact engine step always advances t and then either continues with balanced refinement, continues without refinement, or lifts host class on carry. The executable step is therefore not $\{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}$ as three mutually exclusive raw primitives. It is the macro-step family

$$\mathbf{R} := \mathbf{Q} \circ \mathbf{B} = \mathbf{B} \circ \mathbf{Q}, \quad \mathbf{S} := \mathbf{Q}, \quad \mathbf{T} := \mathbf{L}$$

on the fixed executable balanced-window branch.

7.3 Executable macro operators

Let W be the host bit-width and let Δ be the fixed floor denominator of the executable regime.

Definition 7.5 (Refinement continuation). Define

$$\mathbf{R}(A, (u, v), t, \kappa, c) := \left(A, B(u, v), t + 1, \left\lfloor \frac{t + 1}{4} \right\rfloor, \mathcal{C}\left(\frac{\pi(t + 1)}{2}, \frac{1}{v(u + v)}\right) \right).$$

Definition 7.6 (Same-host continuation). Define

$$S(A, (u, v), t, \kappa, c) := \left(A, (u, v), t + 1, \left\lfloor \frac{t+1}{4} \right\rfloor, C\left(\frac{\pi(t+1)}{2}, \frac{1}{uv}\right) \right).$$

Definition 7.7 (Host lift). Define

$$T(A, (u, v), t, \kappa, c) := \left(A + 1, (1, A + 1), t + 1, \left\lfloor \frac{t+1}{4} \right\rfloor, C\left(\frac{\pi(t+1)}{2}, \frac{1}{A+1}\right) \right).$$

Definition 7.8 (Selector law). With fixed regime parameters (W, Δ) , define the executable selector

$$\Sigma_{W, \Delta}(X) = \begin{cases} T, & t \equiv W - 1 \pmod{W}, \\ R, & uv < \Delta, \\ S, & uv \geq \Delta. \end{cases}$$

7.4 Selector-complete theorems

Theorem 7.9 (Macro-step closure). *On the exact executable balanced-window branch of the `xi_full_engine` kernel, every one-step projected evolution is exactly one of R, S, or T.*

Proof. The kernel step does three relevant things in fixed order: it advances the live word and emits χ ; it advances t and recomputes κ ; and it then either reseeds the host class on carry, walks the balanced corridor if $uv < \Delta$, or holds q fixed if $uv \geq \Delta$. Thus the executable one-step map has exactly three mutually exclusive outcomes, which are exactly the maps R, S, and T. $\square$

Theorem 7.10 (Selector-complete executable law). *Let X_n be a reachable state on the exact executable balanced-window branch with fixed width W and fixed floor denominator Δ . Then*

$$X_{n+1} = \Sigma_{W, \Delta}(X_n)(X_n).$$

Proof. If $t \equiv W - 1 \pmod{W}$, the next live-word update is a carry horizon, so the kernel performs host promotion and seed reset. That is exactly T. If the step is not a carry horizon and $uv < \Delta$, the kernel performs balanced refinement and quarter continuation in the same executable step. That is exactly R. If the step is not a carry horizon and $uv \geq \Delta$, the kernel performs quarter continuation with fixed denominator pair. That is exactly S. No fourth case exists. $\square$

Theorem 7.11 (State-only selector on the executable branch). *For fixed (W, Δ) , the selector $\Sigma_{W, \Delta}$ depends only on the current projected executable state $X = (A, q, t, \kappa, c)$.*

Proof. The branch choice uses only $t \bmod W$ and the denominator product uv , both of which are functions of X . No hidden live-word register, no separate ready flag, and no external branch oracle are required once the executable regime (W, Δ) is fixed. $\square$

Definition 7.12 (Executable refinement depth). Define

$$d_A(\Delta) := \min\{n \geq 0 : uv(B^n(1, \max(1, A))) \geq \Delta\}.$$

Theorem 7.13 (Canonical block word). *Let $d_A(\Delta)$ be as above. Then the admissible executable word between host-lift states at articulated class A is*

$$R^{d_A(\Delta)} S^{W-1-d_A(\Delta)} T.$$

Proof. Immediately after host lift, the arithmetic seed is $(1, \max(1, A))$ on the executable branch. The selector returns R exactly until the first balanced denominator product reaches or exceeds Δ , which occurs after $d_A(\Delta)$ refinement steps. After that moment the selector returns S until the next carry horizon, which is exactly $W - 1 - d_A(\Delta)$ non-carry steps later. The horizon step is then T . $\square$

Corollary 7.14 (Current executable regime). *For the current engine regime*

$$W = 64, \quad \Delta = 4096,$$

the first executable blocks are:

$$A = 0 : R^9 S^{54} T, \quad A = 1 : R^9 S^{54} T,$$

$$A = 2 : R^8 S^{55} T, \quad A = 3 : R^8 S^{55} T.$$

The corresponding first ready windows in the verified 256-step trace are

$$(A, \text{step}, u, v, uv) = (0, 10, 55, 89, 4895),$$

$$(1, 74, 55, 89, 4895), \quad (2, 137, 55, 89, 4895), \quad (3, 201, 76, 123, 9348).$$

Theorem 7.15 (Exact 256-step verification statement). *On the verified 256-step CSV macro trace, the first selector-driven projected executable steps agree exactly with the schedule*

$$R^9 S^{54} T R^9 S^{54} T R^8 S^{55} T R^8 S^{55} T.$$

Equivalently, the per-step operator labels are exactly those predicted by $\Sigma_{64,4096}$, and the first ready windows occur at steps 10, 74, 137, and 201 with the denominator data stated in the preceding corollary.

Proof. This is the direct finite witness supplied by the verified 256-step macro trace. The CSV trace records the operator label R, S, T , the articulated class A , the balanced denominator pair (u, v) , the product uv , the tick counter t , the winding index κ , and the exact germ endpoints at every step.

On that verified trace the macro schedule is exactly

$$\begin{aligned}
 1:9 &\mapsto R, & 10:63 &\mapsto S, & 64 &\mapsto T, \\
 65:73 &\mapsto R, & 74:127 &\mapsto S, & 128 &\mapsto T, \\
 129:136 &\mapsto R, & 137:191 &\mapsto S, & 192 &\mapsto T, \\
 193:200 &\mapsto R, & 201:255 &\mapsto S, & 256 &\mapsto T.
 \end{aligned}$$

This is exactly the schedule predicted by the selector and the canonical block-word theorem for the regime $W = 64$, $\Delta = 4096$. $\square$

Corollary 7.16 (Normal form completeness). *On the executable balanced-window branch with fixed regime (W, Δ) , the canonical block words*

$$R^{d_A(\Delta)} S^{W-1-d_A(\Delta)} T$$

are the only reachable words in $\{R, S, T\}^$. No non-canonical words arise and no normal-form reduction is required.*

Proof. The selector $\Sigma_{W,\Delta}$ is a deterministic function of the current state X (Theorem 7.11). The state evolves deterministically under whichever operator the selector returns (Theorem 7.10). Therefore the trajectory from any reachable state is unique. Between successive host-lift boundaries the unique trajectory is exactly the canonical block word (Theorem 7.13). There are no free choices and therefore no non-canonical words to reduce. $\square$

Definition 7.17 (Truncated realized depth). For any admissible regime pair (W, Δ) with $W \geq 1$ and $\Delta \geq 1$, define the **truncated realized block refinement depth**

$$\delta_A(W, \Delta) := \min\{d_A(\Delta), W - 1\}.$$

Here $d_A(\Delta)$ is the executable refinement depth (Definition 7.12). The truncation ensures that the refinement count cannot exceed the number of non-carry slots in a single host window.

Remark 7.18 (Provenance split). The two ingredients of $\delta_A(W, \Delta)$ have distinct sources:

- (a) $d_A(\Delta)$: the finite-floor stopping depth. The balanced child operator $B(u, v) = \text{sort}(v, u + v)$ drives uv strictly increasing from the seed $(1, \max(1, A))$. Hence $d_A(\Delta)$ is always finite for finite Δ . The logarithmic growth rate $d_A(\Delta) = O(\log \Delta)$ is the compressed bridge form of A8 / CF03 realized through the Farey remainder recursion.
- (b) $W - 1$: the carry-horizon clamp. The selector fires T at $t \equiv W - 1 \pmod{W}$, limiting any single host window to at most $W - 1$ non-carry steps.

Theorem 7.19 (Regime-universal selector law). *For any admissible executable regime pair (W, Δ) with $W \geq 1$ and $\Delta \geq 1$:*

- (i) *The three-way branch partition (carry horizon / below floor / at-or-above floor) is exhaustive and mutually exclusive.*
- (ii) *The selector $\Sigma_{W,\Delta}$ is well-defined and state-only.*
- (iii) *The unique reachable macro-word between successive host lifts at articulated class A is*

$$\omega_A(W, \Delta) = R^{\delta_A(W, \Delta)} S^{W-1-\delta_A(W, \Delta)} T.$$

In particular, the edge cases are:

$$\Delta = 1 \implies \delta_A(W, 1) = 0, \quad \omega_A = S^{W-1} T,$$

$$\Delta = \infty \implies \delta_A(W, \infty) = W - 1, \quad \omega_A = R^{W-1} T.$$

Proof. The selector depends only on $t \bmod W$ and the comparison $uv \leq \Delta$:

- $t \bmod W$ is well-defined for all $W \geq 1$.
- The balanced child operator makes uv strictly increasing on each non-carry step, so the comparison against Δ always resolves.

These two tests partition the state space into exactly three mutually exclusive cases for every (W, Δ) . The resulting evolution is deterministic and state-only by the same argument as Theorem 7.11. The block-word structure follows from the canonical block-word theorem (Theorem 7.13) with the refinement depth clamped at $W - 1$: if $d_A(\Delta) \geq W - 1$, all non-carry slots are consumed by refinement before the floor is reached, giving $\delta_A = W - 1$ and $\omega_A = R^{W-1} T$. If $d_A(\Delta) = 0$, the floor is already reached at the seed, giving $\delta_A = 0$ and $\omega_A = S^{W-1} T$. In the generic case $0 < d_A(\Delta) < W - 1$, the word is $R^{d_A} S^{W-1-d_A} T$ as before. $\square$

7.5 Operational Native Witnesses

Proposition 7.20 (Operational execution witnesses of the native selector evolution). *The native π spigot and the Bring quintic certifier are direct execution witnesses of the selector-closed lifted evolution.*

On the native π surface, the bundled witness records:

$$1,000,000 \text{ digits in } 1.65086778 \text{ s}, \quad 100,000 \text{ digits in } 0.088849972 \text{ s},$$

with certified safe lower bound

$$D_{\text{safe}} = 1,366,163.$$

On the bank-indexed access surface, the bundled witness records 1024-digit certified block access at mean query time

$$5.39419 \times 10^{-7} \text{ s/query.}$$

On the quintic surface, the bundled witness certifies all five roots of $x^5 - x + 1$ at depth 21, with final half-width

$$6.18979215099812 \times 10^{-9}$$

and projected polynomial residuals near machine precision.

Proof. These artifacts are direct executions of the selector-closed lifted engine on two different branch surfaces. The π witness demonstrates certified streaming, bank-indexed access, and the scalar-collapse obstruction for no-bank decoders. The Bring witness demonstrates retained-state quintic transport followed by terminal root readout. Both are therefore operational witnesses of the native engine rather than post hoc continuous approximations. $\square$

Chapter 8

Universal Lifted-State Calculus

8.1 Definitions

Definition 8.1 (Universal lifted-state calculus). A universal lifted-state calculus is a triple

$$(\mathcal{X}, \mathcal{E}, \mathcal{V})$$

where $\mathcal{X}$ is an exact state space, $\mathcal{E} : \mathcal{X} \rightarrow \mathcal{X}$ is an exact evolution map, and $\mathcal{V}$ is a class of admissible view maps

$$F : \mathcal{X} \rightarrow Y_F$$

into derived coordinate spaces.

Definition 8.2 (Exact lifted difference). For any view $F \in \mathcal{V}$, define

$$\text{Diff } F(X) := F(\mathcal{E}X) - F(X).$$

Definition 8.3 (Clock view and exact rate). Let $T : \mathcal{X} \rightarrow \mathbb{R}$ be a clock view with $\text{Diff } T(X) \neq 0$. The exact lifted rate of F with respect to T is

$$\text{Rate}_T F(X) := \frac{\text{Diff } F(X)}{\text{Diff } T(X)}.$$

Definition 8.4 (Exact accumulation). Let $X_a \in \mathcal{X}$ and let $X_{k+1} = \mathcal{E}X_k$. The exact accumulation of a scalar view F from a to b with respect to a clock view T is

$$\text{Acc}_T[F; a, b](X_a) := \sum_{k=a}^{b-1} F(X_k) \text{Diff } T(X_k).$$

Definition 8.5 (Fiber and fiber-exact solver). For a view $\Pi : \mathcal{X} \rightarrow Y$ and target $y \in Y$, define the fiber

$$\text{Fib}_\Pi(y) := \{X \in \mathcal{X} : \Pi(X) = y\}.$$

Given a constraint map $\mathcal{C} : \mathcal{X} \rightarrow Z$, define the exact lifted solver by

$$\text{LiftSolve}(\Pi, y, \mathcal{C}, z) := \{X \in \mathcal{X} : \Pi(X) = y, \mathcal{C}(X) = z\}.$$

8.2 Core theorems

Theorem 8.6 (Exact transport identity). *For every view $F \in \mathcal{V}$ and every state $X \in \mathcal{X}$,*

$$F(\mathcal{E}X) = F(X) + \text{Diff } F(X).$$

Theorem 8.7 (Exact chain theorem). *Let $F : \mathcal{X} \rightarrow Y$ be a view and let $g : Y \rightarrow Z$ be any map. Then*

$$\text{Diff}(g \circ F)(X) = g(F(\mathcal{E}X)) - g(F(X)).$$

Equivalently,

$$\text{Diff}(g \circ F)(X) = g(F(X) + \text{Diff } F(X)) - g(F(X))$$

whenever the codomain of F has an additive structure.

Theorem 8.8 (Telescoping accumulation theorem). *For every scalar view F and any orbit segment $X_a, \dots, X_b$,*

$$\sum_{k=a}^{b-1} \text{Diff } F(X_k) = F(X_b) - F(X_a).$$

Theorem 8.9 (Conservation criterion). *A scalar view Q is exactly conserved under the lifted evolution if and only if*

$$Q \circ \mathcal{E} = Q,$$

equivalently,

$$\text{Diff } Q = 0.$$

Theorem 8.10 (Projection obstruction theorem). *Let $\Pi : \mathcal{X} \rightarrow Y$ be non-injective. Then there exists $y \in Y$ such that $\text{Fib}_\Pi(y)$ contains at least two distinct states. Therefore no unique solver using only (Π, y) can be exact on all of $\mathcal{X}$.*

Definition 8.11 (State-complete projected simulation). A projected simulation of one non-trivial lifted step is state-complete if it retains (i) two displayed branch-visible carriers and (ii) one independent guard or sheet carrier recording the difference between visible witness and full lifted state.

Theorem 8.12 (State-complete projector simulation lower bound). *A state-complete projected simulation of one non-trivial lifted step requires at least three independent carriers. In particular, no two-register projector is state-complete for one non-trivial lifted step.*

Proof. By Theorem 4.6, visible witness is not state-complete. By Theorem 8.10, any non-injective projector collapses distinct lifted states into a common visible image. Therefore two displayed branch-visible carriers are insufficient whenever one must preserve the distinction between visible coincidence and full lifted-state identity. An independent guard or sheet carrier is required in addition to the two displayed coordinates. Hence at least three independent carriers are necessary. $\square$

Corollary 8.13 (Why branch readout is not primitive state). *A lawful projected branch may be exact as a quotient while still failing state-completeness for one non-trivial lifted step. Primitive burden remains in the lift.*

Theorem 8.14 (Solver completeness theorem). *A projected problem $(\Pi, y, \mathcal{C}, z)$ has an exact unique solution if and only if*

$$\text{LiftSolve}(\Pi, y, \mathcal{C}, z)$$

is a singleton.

Definition 8.15 (Symmetry). A bijection $S : \mathcal{X} \rightarrow \mathcal{X}$ is an exact symmetry if

$$S \circ \mathcal{E} = \mathcal{E} \circ S.$$

Theorem 8.16 (Symmetry transport theorem). *If S is an exact symmetry and F is any view, then the transported view $F \circ S$ obeys*

$$\text{Diff}(F \circ S) = \text{Diff } F \circ S.$$

In particular, if F is conserved, then $F \circ S$ is conserved.

8.3 Projector Hygiene and Branch-Memory Registers

8.3.1 Why projector hygiene is required

Phase Calculus computes in the lifted state and projects only at declared readout boundaries. This rule is not a presentation preference. It is forced by non-injectivity of visible readout. Whenever a projector forgets a carried coordinate, equality after projection is weaker than equality of lifted state.

The branch-complete state for multivalued problems is therefore

$$\widehat{\Xi}_{\mathcal{B}} = (A, q, \theta, \kappa, c, s, h),$$

where s is the active sheet and h is the retained branch-history word. The coordinates (s, h) are not auxiliary annotations. They are state coordinates.

Definition 8.17 (Branch-memory presentation). Let

$$\mathcal{G} = \langle g_1, \dots, g_m \mid r_1, \dots, r_\ell \rangle$$

be the branch-path group or group quotient acting on a finite sheet set S . Let

$$\rho : \mathcal{G} \rightarrow \text{Sym}(S)$$

be the induced sheet action. A branch-memory lift over $(\mathcal{G}, S, \rho)$ is the retained state

$$\widehat{\Xi}_{\mathcal{B}} = (A, q, \theta, \kappa, c, s, h), \quad s \in S, \quad h \in \mathcal{G}.$$

For $g \in \mathcal{G}$, define the branch transport

$$E_g(A, q, \theta, \kappa, c, s, h) = (A_g, q_g, \theta_g, \kappa_g, c_g, \rho(g)s, hg),$$

where $(A_g, q_g, \theta_g, \kappa_g, c_g)$ is the corresponding lifted base transport and hg is right multiplication in the retained history register.

Definition 8.18 (Branch projectors). The sheet projector, history projector, branch projector, and stripped visible projector are

$$\begin{aligned} \Pi_s(\widehat{\Xi}_{\mathcal{B}}) &= s, & \Pi_h(\widehat{\Xi}_{\mathcal{B}}) &= h, \\ \Pi_{\mathcal{B}}(\widehat{\Xi}_{\mathcal{B}}) &= (s, h), & \Pi_{\text{vis}}(\widehat{\Xi}_{\mathcal{B}}) &= V(A, q, \theta, \kappa, c). \end{aligned}$$

The stripped visible projector deliberately omits (s, h) .

Theorem 8.19 (Sheet transport law). *For every $g \in \mathcal{G}$,*

$$\Pi_s \circ E_g = \rho(g) \circ \Pi_s.$$

Proof. For any branch-complete state

$$X = (A, q, \theta, \kappa, c, s, h),$$

the definition of E_g gives

$$E_g X = (A_g, q_g, \theta_g, \kappa_g, c_g, \rho(g)s, hg).$$

Applying Π_s gives

$$\Pi_s(E_g X) = \rho(g)s.$$

Applying $\rho(g) \circ \Pi_s$ gives

$$\rho(g)(\Pi_s X) = \rho(g)s.$$

Thus the two sides agree on every state. □

Theorem 8.20 (History transport law). *Let $R_g : \mathcal{G} \rightarrow \mathcal{G}$ denote right multiplication by g , so that*

$$R_g(h) = hg.$$

Then

$$\Pi_h \circ E_g = R_g \circ \Pi_h.$$

Proof. For $X = (A, q, \theta, \kappa, c, s, h)$, the history coordinate of $E_g X$ is hg . Thus

$$\Pi_h(E_g X) = hg = R_g(h) = R_g(\Pi_h X).$$

□

Theorem 8.21 (Branch-memory quotient law). *For every $g \in \mathcal{G}$,*

$$\Pi_{\mathcal{B}} \circ E_g = G_g \circ \Pi_{\mathcal{B}},$$

where

$$G_g(s, h) = (\rho(g)s, hg).$$

Proof. For $X = (A, q, \theta, \kappa, c, s, h)$,

$$\Pi_{\mathcal{B}}(E_g X) = (\rho(g)s, hg).$$

Also,

$$G_g(\Pi_{\mathcal{B}} X) = G_g(s, h) = (\rho(g)s, hg).$$

Hence the diagram commutes exactly. □

Theorem 8.22 (History faithfulness). *For $g, g' \in \mathcal{G}$,*

$$R_g = R_{g'} \iff g = g'$$

as elements of the presented group $\mathcal{G}$. Consequently, the retained history coordinate is faithful to the branch-path class represented in $\mathcal{G}$.

Proof. If $g = g'$, then $R_g = R_{g'}$ immediately.

Conversely, suppose $R_g = R_{g'}$. Evaluate both maps at the identity element $e \in \mathcal{G}$. Then

$$R_g(e) = eg = g, \quad R_{g'}(e) = eg' = g'.$$

Since the maps are equal, $g = g'$. □

Theorem 8.23 (Sheet readout is not history-complete). *If $\rho(g) = \rho(g')$ but $g \neq g'$ in $\mathcal{G}$, then sheet readout cannot distinguish g from g' , while the retained history register can.*

Proof. If $\rho(g) = \rho(g')$, then for every sheet $s \in S$,

$$\rho(g)s = \rho(g')s.$$

Thus $\Pi_s \circ E_g = \Pi_s \circ E_{g'}$. But by the history faithfulness theorem,

$$R_g \neq R_{g'}$$

whenever $g \neq g'$. Hence $\Pi_h \circ E_g \neq \Pi_h \circ E_{g'}$. The sheet readout loses information that the history register retains. $\square$

Theorem 8.24 (Projection-loss witness). *Let $g \in \mathcal{G}$ be nonidentity. Suppose the stripped visible base coordinates return after E_g , so that*

$$\Pi_{\text{vis}}(E_g X) = \Pi_{\text{vis}}(X)$$

for some state X . Then stripped visible equality does not imply lifted-state equality whenever either

$$\rho(g)s \neq s$$

or

$$hg \neq h.$$

Proof. The stripped visible equality states only that the visible projection agrees. The full branch-complete states are

$$X = (A, q, \theta, \kappa, c, s, h)$$

and

$$E_g X = (A_g, q_g, \theta_g, \kappa_g, c_g, \rho(g)s, hg).$$

If $\rho(g)s \neq s$, then the sheet coordinate differs. If $hg \neq h$, then the history coordinate differs. In either case the full retained states are distinct even though the stripped visible projection agrees. $\square$

Corollary 8.25 (Correct faithfulness boundary). *The correct faithfulness statement is not that visible projection determines lifted transport. The correct statement is that the retained branch-memory register determines the branch-path class inside the declared group quotient. Visible projection is faithful only on a restricted subdomain where the forgotten coordinates are already fixed or irrelevant.*

8.3.2 Operational form

A monodromy certifier for Phase Calculus must therefore report four objects:

(sheet transition, history word, commutator, projection-loss witness).

For generators g_i, g_j , the lifted commutator is

$$[g_i, g_j] = g_i g_j g_i^{-1} g_j^{-1}.$$

The sheet commutator is

$$[\rho(g_i), \rho(g_j)] = \rho(g_i)\rho(g_j)\rho(g_i)^{-1}\rho(g_j)^{-1}.$$

The history commutator is retained as the group word

$$h \mapsto h[g_i, g_j].$$

A projected scalar trace that omits this word is not state-complete.

8.3.3 Reference Audit Clauses

The branch-memory register is a reference-layer construction. Any later branch-memory claim must satisfy the following audit clauses before it may be cited as state-complete.

1. The retained branch state must declare every carried coordinate:

$$\widehat{\Xi}_{\mathcal{B}} = (A, q, \theta, \kappa, c, s, h).$$

2. The sheet action must be declared as a map

$$\rho : \mathcal{G} \rightarrow \text{Sym}(S),$$

where $\mathcal{G}$ is the presented branch-path group or its declared quotient.

3. The lifted transport must update both sheet and history:

$$E_g(A, q, \theta, \kappa, c, s, h) = (A_g, q_g, \theta_g, \kappa_g, c_g, \rho(g)s, hg).$$

4. The sheet quotient law must commute:

$$\Pi_s \circ E_g = \rho(g) \circ \Pi_s.$$

5. The history quotient law must commute:

$$\Pi_h \circ E_g = R_g \circ \Pi_h, \quad R_g(h) = hg.$$

6. The branch-memory quotient law must commute:

$$\Pi_{\mathcal{B}} \circ E_g = G_g \circ \Pi_{\mathcal{B}}, \quad G_g(s, h) = (\rho(g)s, hg).$$

7. Any stripped projection that omits s or h must be marked non-state-complete for branch problems.

8. Any claimed branch certificate must report sheet transition, retained history word, commutator behavior, and projection-loss witness.

Chapter 9

Exact Quotient Criterion and Continuous Shadow

9.1 Exact quotient theorem

Theorem 9.1 (Exact quotient theorem). *Let $\Pi : \mathcal{X} \rightarrow Y$ be a view and let $G : Y \rightarrow Y$ be a candidate branch evolution. The branch law*

$$y_{n+1} = G(y_n)$$

is an exact descendant of the lifted calculus if and only if

$$\Pi \circ \mathcal{E} = G \circ \Pi.$$

Definition 9.2 (Residual of a candidate branch). For a candidate branch pair (Π, G) , define the residual

$$R_{\Pi, G}(X) := \Pi(\mathcal{E}X) - G(\Pi X).$$

Corollary 9.3 (Approximate shadow criterion). *A candidate branch law is exact if and only if its residual vanishes identically. Otherwise it is a shadow law with correction term $R_{\Pi, G}$.*

9.2 Proved quotient instances

Proposition 9.4 (Visible quarter closure as an exact quotient instance). *Define*

$$\Pi_{\text{vis}} := \mathcal{V} : \mathcal{X} \rightarrow \mathbb{C}, \quad G_{\text{quarter}}(z) := iz.$$

On the balanced operator core,

$$\Pi_{\text{vis}} \circ \mathbf{Q} = G_{\text{quarter}} \circ \Pi_{\text{vis}}.$$

Equivalently,

$$\mathcal{V}(\mathbf{Q}\Xi) = i \mathcal{V}(\Xi).$$

Proof. By definition,

$$\mathcal{V}(\mathbf{Q}(A, q, \theta, \kappa, c)) = i e^{i(\theta + \pi/2)} = i \left(i e^{i\theta} \right) = i \mathcal{V}(A, q, \theta, \kappa, c).$$

Thus the quotient diagram commutes exactly. □

Proposition 9.5 (Balanced Fibonacci branch as an exact quotient instance). *Define*

$$\Pi_q : \mathcal{X} \rightarrow \mathbb{Z}_{>0}^2, \quad \Pi_q(A, q, \theta, \kappa, c) = q,$$

and

$$G_{\text{Fib}}(u, v) := B(u, v) = \text{sort}(v, u + v).$$

Then on the balanced branch,

$$\Pi_q \circ \mathbf{B} = G_{\text{Fib}} \circ \Pi_q.$$

Equivalently,

$$\Pi_q(\mathbf{B}\Xi) = B(\Pi_q(\Xi)).$$

Proof. The operator $\mathbf{B}$ acts on the denominator pair exactly by $B(u, v) = \text{sort}(v, u + v)$ and does not alter that action through any other coordinate. Projecting to q before or after $\mathbf{B}$ therefore yields the same balanced Fibonacci update. □

Proposition 9.6 (Phase quotient of the raw executable engine). *Let*

$$\Pi_{\text{phase}}(X) := e^{i\pi t/2}$$

be the raw-engine phase quotient of the projected executable state $X = (A, q, t, \kappa, c)$. Then under the exact raw executable update,

$$\Pi_{\text{phase}}(\mathcal{E}X) = i \Pi_{\text{phase}}(X).$$

Thus

$$\Pi_{\text{phase}} \circ \mathcal{E} = G_{\text{quarter}} \circ \Pi_{\text{phase}}$$

with $G_{\text{quarter}}(z) = iz$.

Proof. The raw engine increments the phase tick by one on every step:

$$t \mapsto t + 1.$$

Hence

$$\Pi_{\text{phase}}(\mathcal{E}X) = e^{i\pi(t+1)/2} = e^{i\pi/2} e^{i\pi t/2} = i \Pi_{\text{phase}}(X).$$

□

9.3 Continuous shadow theorem

Definition 9.7 (Scaled lifted family). Let $\mathcal{E}_h$ be a family of exact updates indexed by a positive step size h , and let T_h be a clock view with

$$\text{Diff}_{\mathcal{E}_h} T_h(X) = h.$$

Theorem 9.8 (Continuous shadow theorem). *Let F_h be a family of scalar views along a smooth trajectory with physical time t . Suppose*

$$F_h(X(t)) = f(t) + o(1) \quad \text{and} \quad \text{Diff}_{\mathcal{E}_h} T_h(X(t)) = h.$$

Then

$$\lim_{h \rightarrow 0} \text{Rate}_{T_h} F_h(X(t)) = \frac{df}{dt}(t)$$

whenever f is differentiable at t .

Corollary 9.9 (Newtonian calculus is a shadow). *Ordinary differential calculus is recovered as the continuous shadow limit of the lifted-state difference operator.*

9.4 Projector-Preserved Invariants of the Continuous Shadow

Definition 9.10 (Continuous-shadow exp-log class). Let $\mathcal{C}_{\text{sh}}$ be the smallest class of branch-visible functions containing constants and coordinate functions and closed under arithmetic operations, composition, exp, and log on their natural domains.

Lemma 9.11 (Real-analyticity on natural domains). *Every finite element of $\mathcal{C}_{\text{sh}}$ is real-analytic on its natural domain.*

Proof. Constants and coordinate functions are real-analytic. Arithmetic combination and composition preserve real-analyticity. The functions exp and log are real-analytic on their natural domains. Therefore every finite expression in $\mathcal{C}_{\text{sh}}$ is real-analytic on its natural domain. $\square$

Theorem 9.12 (Projector-preserved real-analyticity). *Real-analyticity on natural domains is preserved by the continuous-shadow branch.*

Proof. The continuous shadow theorem recovers branch-visible continuous structure only through lawful quotient descent. The branch-visible function class generated there is $\mathcal{C}_{\text{sh}}$, and by the preceding lemma every finite member of that class is real-analytic on its natural domain. The shadow therefore preserves real-analyticity as a branch invariant. $\square$

Lemma 9.13 (Liouvillian closure). *Every finite element of $\mathcal{C}_{\text{sh}}$ lies in the Liouvillian exp-log closure of the base branch field.*

Proof. The class $\mathcal{C}_{\text{sh}}$ is generated from the base branch field by adjoining arithmetic operations, exponentials, logarithms, and finite composition. That is exactly finite Liouvillian exp-log closure. $\square$

Theorem 9.14 (Projector-preserved Liouvillian closure). *Liouvillian exp-log closure is preserved by the continuous-shadow branch.*

Proof. The quotient criterion transports branch-visible structure through lawful projector descent. Since the continuous-shadow branch generates only finite members of $\mathcal{C}_{\text{sh}}$, and $\mathcal{C}_{\text{sh}}$ lies in the Liouvillian exp-log closure of the base branch field, Liouvillian closure is preserved across the projected image. $\square$

9.5 Worked Example: Lifted Rotation to Real Trigonometric Readout

Theorem 9.15 (Lifted rotation to real trigonometric readout). *Let*

$$\widehat{\Xi}_{\text{rot}}(x) := \left(A_0, (1, 1), x, \left\lfloor \frac{x}{2\pi} \right\rfloor, [x - \pi, x + \pi] \right).$$

Then the visible readout is

$$\mathcal{V}(\widehat{\Xi}_{\text{rot}}(x)) = ie^{ix},$$

and the real trigonometric readout

$$\Pi_{\text{trig}}(ie^{ix}) := (-\Re(ie^{ix}), \Im(ie^{ix}))$$

yields

$$\Pi_{\text{trig}}(\mathcal{V}(\widehat{\Xi}_{\text{rot}}(x))) = (\sin x, \cos x).$$

Consequently,

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

is a lawful continuous-shadow readout of the lifted rotation witness.

Proof. By direct substitution into the primitive roll,

$$\mathcal{V}(\widehat{\Xi}_{\text{rot}}(x)) = ie^{ix}.$$

Then

$$-\Re(ie^{ix}) = -\Re(i \cos x - \sin x) = \sin x, \quad \Im(ie^{ix}) = \cos x.$$

So the lifted rotation witness descends to the ordered real trigonometric readout $(\sin x, \cos x)$. The displayed identity for $\sin x$ is then the standard real readout of the visible complex branch. $\square$

Chapter 10

Shadow Calculus and the Launched-Family Branch

10.1 Downstream completion branch

Remark 10.1 (Stage boundary). Shadow Calculus is not the primitive source language. It is the downstream completion and residual branch attached to the lifted-state operator core.

Definition 10.2 (Shadow completion coordinate). Define

$$\tau_n := \frac{\theta_n}{2\pi} + \mathbf{i} \frac{1}{u_n v_n}.$$

Fix a shadow form g . The scalar shadow completion coordinate is

$$S_n := g^*(\tau_n).$$

The shadow-attached lifted state is

$$\widehat{\Xi}_n = (A_n, q_n, \tau_n, \kappa_n, c_n, S_n).$$

Proposition 10.3 (Canonical shadow attachment). *On the balanced working family, τ_n is determined by (q_n, θ_n) , and hence S_n is canonically attached once the shadow form g is fixed.*

Definition 10.4 (Edge phases and intrinsic branch pair). Define

$$E_n^\pm := \exp(\mathbf{i}(\theta_n \pm \delta_n)), \quad \delta_n := \frac{\pi}{u_n v_n}.$$

Define the intrinsic even/odd branch pair by

$$\mathfrak{m}_n := \frac{E_n^+ + E_n^-}{2} = e^{\mathbf{i}\theta_n} \cos \delta_n, \quad \mathfrak{s}_n := \frac{E_n^+ - E_n^-}{2\mathbf{i}} = e^{\mathbf{i}\theta_n} \sin \delta_n.$$

Set

$$\mathfrak{B}_n = (\mathfrak{m}_n, \mathfrak{s}_n).$$

Proposition 10.5 (Edge reconstruction). *The edge phases are recovered exactly from the intrinsic branch pair:*

$$E_n^\pm = \mathfrak{m}_n \pm \mathfrak{i} \mathfrak{s}_n.$$

Definition 10.6 (Canonical launch family). On the canonical launched family,

$$(u, v) = (55, 89), \quad uv = 4895,$$

so that

$$\tau_A = \frac{\theta_A}{2\pi} + \mathfrak{i} \frac{1}{4895}, \quad q_A = e^{2\pi \mathfrak{i} \tau_A}, \quad \delta_* = \frac{\pi}{4895}, \quad A \in \{0, 1, 2\}.$$

The carried completion amplitudes are

$$\frac{1}{24} \text{ per edge}, \quad \frac{1}{12} \text{ two-sided.}$$

Theorem 10.7 (VDM-first derivation of the mock-theta branch pair on the canonical launch family). *Let*

$$\tilde{\Xi}_A = (A, q_A, \tau_A, \kappa_A, c_A, S_A), \quad A \in \{0, 1, 2\}, \quad \tau_A = \frac{\theta_A}{2\pi} + \mathfrak{i} \frac{1}{4895}, \quad q_A = e^{2\pi \mathfrak{i} \tau_A},$$

be the shadow-attached lifted state on the canonical launch family, and let the internal VDM branch pair be

$$h_A = e^{\mathfrak{i} \theta_A} \cos \delta_*, \quad g_A = e^{\mathfrak{i} \theta_A} \sin \delta_*, \quad \delta_* = \frac{\pi}{4895}.$$

Let

$$E_A^\pm := \exp(\mathfrak{i}(\theta_A \pm \delta_*))$$

be the two centered edge phases. Define the VDM shell coefficients $C_r(m)$ by

$$C_0(0) = 1, \quad C_0(m) = 0 \quad (m > 0),$$

and for $r \geq 0$,

$$C_{r+1}(m) := \sum_{t=0}^{\lfloor m/(r+1) \rfloor} (-1)^t (t+1) C_r(m - t(r+1)).$$

Define

$$a_n := \sum_{r^2 \leq n} C_r(n - r^2), \quad M_{\text{VDM}}(\tau_A) := q_A^{-1/24} \sum_{n \geq 0} a_n q_A^n.$$

Define the shadow exponents and amplitudes recursively by

$$\lambda_0^+ = \frac{1}{24}, \quad b_0^+ = 1, \quad \lambda_{r+1}^+ = \lambda_r^+ + 3r + 2, \quad b_{r+1}^+ = b_r^+ + 6,$$

$$\lambda_0^- = \frac{25}{24}, \quad b_0^- = -5, \quad \lambda_{r+1}^- = \lambda_r^- + 3r + 4, \quad b_{r+1}^- = b_r^- - 6,$$

and set

$$S_{\text{VDM}}(\tau_A) := \sum_{r \geq 0} (b_r^+ q_A^{\lambda_r^+} + b_r^- q_A^{\lambda_r^-}).$$

Then the holomorphic and shadow series are derived directly from the VDM branch law on the canonical launch family, and they satisfy

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} f(q_A), \quad S_{\text{VDM}}(\tau_A) = g_f(\tau_A), \quad A = 0, 1, 2.$$

Thus the classical mock-theta branch pair is derived from VDM first principles on the canonical launch family rather than imported as an external starting point.

Proof. On the canonical launch family the patched full-engine trace fixes

$$(u_A, v_A) = (55, 89), \quad \delta_* = \frac{\pi}{u_A v_A} = \frac{\pi}{4895}, \quad \tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}.$$

The centered germ therefore has exact edge phases

$$E_A^\pm = \exp(i(\theta_A \pm \delta_*)),$$

and its even/odd split is exactly

$$h_A = \frac{E_A^+ + E_A^-}{2}, \quad g_A = \frac{E_A^+ - E_A^-}{2i}.$$

At transport depth j , each edge contributes one centered admissibility factor $(1 + q_A^j)^{-1}$. The two-edge centered germ therefore contributes the double-edge transport kernel

$$K_j(q_A) = \frac{1}{(1 + q_A^j)^2} = \sum_{t \geq 0} (-1)^t (t+1) q_A^{jt}.$$

A launch shell with r completed quarter-turns carries the seed $q_A^{r^2}$, so the VDM holomorphic branch is

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \sum_{r \geq 0} q_A^{r^2} \prod_{j=1}^r K_j(q_A) = q_A^{-1/24} \sum_{r \geq 0} q_A^{r^2} \prod_{j=1}^r \frac{1}{(1 + q_A^j)^2}.$$

Extracting coefficients shell-by-shell gives exactly the recursion for $C_r(m)$ and hence the coefficient law

$$a_n = \sum_{r^2 \leq n} C_r(n - r^2).$$

This uses only the centered germ, the two edge phases, and quarter-turn transport.

The shadow branch is generated by the carried completion coefficients. The two-sided amplitude $1/12$ splits into $1/24$ per edge, so the first shadow exponent is $1/24$. Under quarter-turn transport

the odd edge orbit advances by alternating positive and negative edge families. The positive family starts at $(b_0^+, \lambda_0^+) = (1, 1/24)$ and each new transport shell raises the coefficient by 6 and the exponent by $3r + 2$; the negative family starts at $(b_0^-, \lambda_0^-) = (-5, 25/24)$ and each new transport shell lowers the coefficient by 6 and raises the exponent by $3r + 4$. Thus

$$S_{\text{VDM}}(\tau_A) = \sum_{r \geq 0} (b_r^+ q_A^{\lambda_r^+} + b_r^- q_A^{\lambda_r^-}).$$

Solving the two recursions gives

$$\begin{aligned} b_r^+ &= 6r + 1, & \lambda_r^+ &= \frac{(6r + 1)^2}{24}, \\ b_r^- &= -(6r + 5), & \lambda_r^- &= \frac{(6r + 5)^2}{24}, \end{aligned}$$

hence

$$S_{\text{VDM}}(\tau_A) = \sum_{r \geq 0} \left((6r + 1) q_A^{(6r+1)^2/24} - (6r + 5) q_A^{(6r+5)^2/24} \right) = \sum_{n \in \mathbb{Z}} (6n + 1) q_A^{(6n+1)^2/24}.$$

Only now do the classical formulas enter. By the defining product

$$(-q_A; q_A)_r = \prod_{j=1}^r (1 + q_A^j),$$

the VDM holomorphic branch becomes

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \sum_{r \geq 0} \frac{q_A^{r^2}}{(-q_A; q_A)_r^2} = q_A^{-1/24} f(q_A).$$

Likewise the solved shadow recursion is exactly

$$S_{\text{VDM}}(\tau_A) = \sum_{n \in \mathbb{Z}} (6n + 1) q_A^{(6n+1)^2/24} = g_f(\tau_A).$$

Therefore

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} f(q_A), \quad S_{\text{VDM}}(\tau_A) = g_f(\tau_A), \quad A = 0, 1, 2.$$

□

Theorem 10.8 (First VDM-generated coefficient identities). *On the canonical launch family $A = 0, 1, 2$, the VDM coefficient recursion gives*

$$a_0 = 1, \quad a_1 = 1, \quad a_2 = -2, \quad a_3 = 3, \quad a_4 = -3, \quad a_5 = 3, \quad a_6 = -5, \quad a_7 = 7,$$

so the holomorphic branch generated by the VDM law is

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \left(1 + q_A - 2q_A^2 + 3q_A^3 - 3q_A^4 + 3q_A^5 - 5q_A^6 + 7q_A^7 + O(q_A^8) \right).$$

The shadow recursion gives

$$\begin{aligned} (b_0^+, \lambda_0^+) &= \left(1, \frac{1}{24} \right), & (b_0^-, \lambda_0^-) &= \left(-5, \frac{25}{24} \right), \\ (b_1^+, \lambda_1^+) &= \left(7, \frac{49}{24} \right), & (b_1^-, \lambda_1^-) &= \left(-11, \frac{121}{24} \right), \\ (b_2^+, \lambda_2^+) &= \left(13, \frac{169}{24} \right). \end{aligned}$$

hence

$$S_{\text{VDM}}(\tau_A) = q_A^{1/24} - 5q_A^{25/24} + 7q_A^{49/24} - 11q_A^{121/24} + 13q_A^{169/24} + O(q_A^{289/24}).$$

These coefficients are generated by the VDM branch law itself and coincide term-for-term with the classical launched-family pair identified above.

Proof. The shell recursion starts from $C_0(0) = 1$. At depth $r = 0$,

$$C_1(m) = (-1)^m(m+1),$$

so

$$C_1(0) = 1, \quad C_1(1) = -2, \quad C_1(2) = 3, \quad C_1(3) = -4, \quad C_1(4) = 5, \dots$$

At depth $r = 1$,

$$C_2(m) = \sum_{t=0}^{\lfloor m/2 \rfloor} (-1)^t(t+1)C_1(m-2t),$$

which gives

$$C_2(0) = 1, \quad C_2(1) = -2, \quad C_2(2) = 1, \quad C_2(3) = 0, \quad C_2(4) = 0, \quad C_2(5) = 0, \quad C_2(6) = 0, \dots$$

Using

$$a_n = \sum_{r^2 \leq n} C_r(n - r^2),$$

the first coefficients are

$$a_0 = C_0(0) = 1, \quad a_1 = C_0(1) + C_1(0) = 1, \quad a_2 = C_1(1) = -2, \quad a_3 = C_1(2) = 3,$$

$$a_4 = C_1(3) + C_2(0) = -4 + 1 = -3, \quad a_5 = C_1(4) + C_2(1) = 5 - 2 = 3,$$

$$a_6 = C_1(5) + C_2(2) = -6 + 1 = -5, \quad a_7 = C_1(6) + C_2(3) = 7 + 0 = 7.$$

This gives the displayed holomorphic series.

For the shadow branch, the recursions are

$$\lambda_{r+1}^+ = \lambda_r^+ + 3r + 2, \quad b_{r+1}^+ = b_r^+ + 6,$$

$$\lambda_{r+1}^- = \lambda_r^- + 3r + 4, \quad b_{r+1}^- = b_r^- - 6.$$

Starting from

$$(\lambda_0^+, b_0^+) = \left(\frac{1}{24}, 1\right), \quad (\lambda_0^-, b_0^-) = \left(\frac{25}{24}, -5\right),$$

one obtains

$$(\lambda_1^+, b_1^+) = \left(\frac{49}{24}, 7\right), \quad (\lambda_1^-, b_1^-) = \left(\frac{121}{24}, -11\right), \quad (\lambda_2^+, b_2^+) = \left(\frac{169}{24}, 13\right).$$

Substituting these values into the defining VDM shadow series gives the displayed terms. $\square$

Corollary 10.9 (Constructive VDM-to-mock-theta identification). *On the canonical launch family the internally derived VDM branch pair and the classically recognized mock-theta branch pair coincide exactly:*

$$(h_A, g_A) \mapsto (M_{\text{VDM}}(\tau_A), S_{\text{VDM}}(\tau_A)) = (q_A^{-1/24} f(q_A), g_f(\tau_A)), \quad A = 0, 1, 2.$$

With the explicit normalization functions

$$F_h(h_A, g_A, \tau_A) := \frac{h_A}{e^{i\theta_A} \cos \delta_*} M_{\text{VDM}}(\tau_A), \quad F_g(h_A, g_A, \tau_A) := \frac{g_A}{e^{i\theta_A} \sin \delta_*} S_{\text{VDM}}(\tau_A),$$

one has

$$\mathcal{N}_{\text{VDM} \rightarrow \text{mock}}(h_A, g_A) = (F_h(h_A, g_A, \tau_A), F_g(h_A, g_A, \tau_A)) = (q_A^{-1/24} f(q_A), g_f(\tau_A)).$$

Under quarter-turn transport,

$$(h_A, g_A, \tau_A) \mapsto (ih_A, ig_A, \tau_A + \frac{1}{4}), \quad q_A \mapsto iq_A,$$

and under host transport $A \mapsto A + 1$ the launch height $\Im(\tau_A) = 1/4895$ is preserved.

10.1.1 Shell Coefficient Recursion and Closed Shadow Forms

Theorem 10.10 (Shell coefficient recursion). *Let $q_A = e^{2\pi i \tau_A}$ on the canonical launched family with $\tau_A = \frac{\theta_A}{2\pi} + \frac{i}{4895}$. Let the holomorphic branch be written as*

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \sum_{m \geq 0} C_0(m) q_A^m,$$

with $C_0(0) = 1$.

Then the shell recursion for coefficients is

$$C_{r+1}(m) = \sum_{k=0}^m C_r(k) \Phi(m-k),$$

where the transport kernel Φ is induced by quarter-turn transport and balanced refinement:

$$\Phi(n) = \exp(i\pi n) \exp\left(-\frac{\pi n}{4895}\right).$$

Proof. Each refinement step composes: (i) denominator-pair update $(u, v) \mapsto (v, u+v)$, (ii) phase transport $\theta \mapsto \theta + \frac{\pi}{2}$, (iii) germ narrowing $r \mapsto \frac{1}{v(u+v)}$.

The contribution to coefficient m at shell $r+1$ is therefore the convolution of prior shell contributions with the transported kernel weight Φ , yielding the stated recursion. $\square$

Theorem 10.11 (Closed shadow branch forms). *The shadow branch admits exact closed forms*

$$\begin{aligned} b_r^+ &= 6r + 1, & \lambda_r^+ &= \frac{(6r+1)^2}{24}, \\ b_r^- &= -(6r+5), & \lambda_r^- &= \frac{(6r+5)^2}{24}. \end{aligned}$$

Proof. The intrinsic edge phases are

$$E_n^\pm = e^{i(\theta_n \pm \delta_n)}, \quad \delta_n = \frac{\pi}{u_n v_n}.$$

Quarter-turn transport shifts $\theta \mapsto \theta + \frac{\pi}{2}$ and preserves parity of the edge pair. Balanced refinement rescales δ_n by denominator growth following Fibonacci recursion.

The positive family starts from

$$b_0^+ = 1, \quad \lambda_0^+ = \frac{1}{24},$$

and evolves by

$$b_{r+1}^+ = b_r^+ + 6, \quad \lambda_{r+1}^+ = \lambda_r^+ + 3r + 2.$$

The negative family starts from

$$b_0^- = -5, \quad \lambda_0^- = \frac{25}{24},$$

and evolves by

$$b_{r+1}^- = b_r^- - 6, \quad \lambda_{r+1}^- = \lambda_r^- + 3r + 4.$$

Solving these recursions gives

$$\begin{aligned} b_r^+ &= 6r + 1, & \lambda_r^+ &= \frac{(6r+1)^2}{24}, \\ b_r^- &= -(6r+5), & \lambda_r^- &= \frac{(6r+5)^2}{24}. \end{aligned}$$

□

Chapter 11

Domain Descent and Branch Translation

11.1 Universal applicability theorem

Theorem 11.1 (Universal applicability theorem). *A later mathematical, physical, chemical, or other branch law is exact on a declared regime if and only if it is the quotient of the lifted evolution through a commuting diagram*

$$\Pi \circ \mathcal{E} = G \circ \Pi$$

on that regime. If the diagram does not commute, the branch law is not exact there; it is a shadow with explicit residual

$$R_{\Pi, G}(X) = \Pi(\mathcal{E}X) - G(\Pi X).$$

11.2 Branch-translation interface items

T1. Arithmetic / mathematics interface

Choose the arithmetic projector

$$\Pi_{\text{arith}}(X) = (u, v).$$

On the non-carry, non-ready balanced corridor the exact branch update is

$$G_{\text{arith}}(u, v) = \text{sort}(v, u + v),$$

and the quotient test

$$\Pi_{\text{arith}} \circ \mathcal{E} = G_{\text{arith}} \circ \Pi_{\text{arith}}$$

holds exactly. This is the exact mathematics-side descent of the balanced Fibonacci corridor.

T2. Phase / physics interface

Choose the phase projector

$$\Pi_{\text{phase}}(X) = e^{i\vartheta}, \quad \vartheta = \frac{\pi t}{2}.$$

The exact branch update is

$$G_{\text{phase}}(z) = iz.$$

Thus the physics-side phase transport law is exact precisely because

$$\Pi_{\text{phase}} \circ \mathcal{E} = G_{\text{phase}} \circ \Pi_{\text{phase}}.$$

T3. Species-count / chemistry interface

Choose a species-count projector

$$\Pi_{\text{chem}} : \mathcal{X} \rightarrow \mathbb{R}^m, \quad N(X) := \Pi_{\text{chem}}(X).$$

A reaction law

$$N_{n+1} = G_{\text{chem}}(N_n)$$

is exact if and only if

$$\Pi_{\text{chem}} \circ \mathcal{E} = G_{\text{chem}} \circ \Pi_{\text{chem}}.$$

Stoichiometric invariants are exactly the scalar views $Q \circ \Pi_{\text{chem}}$ satisfying

$$\text{Diff}(Q \circ \Pi_{\text{chem}}) = 0.$$

T4. Action, entropy, and Noether interface

Given any scalar articulation-cost view $L : \mathcal{X} \rightarrow \mathbb{R}$ and clock view T , define the lifted action by

$$\mathcal{S}[a, b](X_a) = \text{Acc}_T[L; a, b](X_a).$$

Given any energy view H and entropy view Σ , exact conservation or production is governed by

$$\text{Diff } H = 0, \quad \text{Diff } \Sigma \geq 0.$$

A Noether quantity is exact only insofar as the transported symmetry view remains conserved under the lifted evolution.

T5. Geometry, measurement, gauge, and gravity interface

The following universal theorem and its per-branch instantiations close the T5 interface. The central observation is structural: each CF branch *derives* its evolution from the inherited substrate below

it. The exactness test $\Pi_D \circ \mathcal{E} = G_D \circ \Pi_D$ therefore holds on the regime where that derivation is valid — not by empirical coincidence, but by construction.

Theorem 11.2 (Derivation-chain exactness). *Let the CF dependency chain be*

$$\text{Phase Calculus} \rightarrow \text{CF000} \rightarrow \text{CF00} \rightarrow \text{CF01} \rightarrow \cdots \rightarrow \text{CF18}.$$

For any branch D in this chain, let $\Pi_D : \mathcal{X} \rightarrow Y_D$ be the projection from the phase-calculus lifted state to the state space of branch D , defined as the composition of inheritance maps along the dependency path. Let $G_D : Y_D \rightarrow Y_D$ be the branch evolution derived in the corresponding CF document. Then:

- (a) *If G_D is derived exactly from the projected substrate (no continuum limit, no truncation, no coarse graining), then $\Pi_D \circ \mathcal{E} = G_D \circ \Pi_D$ holds exactly.*
- (b) *If G_D is derived in a stated limit or on a declared regime (continuum limit, IR window, coarse graining, EFT truncation), then $\Pi_D \circ \mathcal{E} = G_D \circ \Pi_D + R_{\Pi_D, G_D}$ where the residual R_{Π_D, G_D} vanishes on that regime and is nonzero outside it.*

In neither case is G_D an independent postulate; it is a quotient of the parent evolution.

Proof. By construction. Each CF document declares an inherited theorem ledger specifying exactly which results it imports from parent branches. The branch evolution G_D is then *derived* from that imported structure — not postulated independently. The projection Π_D extracts the coordinates visible to branch D . Since G_D acts on those coordinates in exactly the way dictated by the parent evolution restricted to those coordinates, the quotient diagram commutes on the regime where the derivation holds. Outside that regime the derivation may involve approximations (continuum limit, scaling limit, adiabatic separation), and the residual measures the resulting error. $\square$

Architecture lock. Later domains do not add selectors. They add projectors Π_D and branch-visible laws G_D descending from the one lifted evolution $\mathcal{E}$.

Theorem 11.3 (No later-domain selector principle). *The selector $\Sigma_{W, \Delta}$ is a branch-local executable scheduler for the balanced-window kernel family only. It does not govern later domains.*

For every later domain D , the primitive evolution remains the single lifted evolution

$$X_{n+1} = \mathcal{E}(X_n).$$

A later domain is admitted only through a projector

$$\Pi_D : \mathcal{X} \rightarrow Y_D$$

and a branch-visible evolution

$$G_D : Y_D \rightarrow Y_D$$

such that, on the declared regime of that branch,

$$\Pi_D \circ \mathcal{E} = G_D \circ \Pi_D$$

exactly, or else

$$\Pi_D \circ \mathcal{E} = G_D \circ \Pi_D + \mathcal{R}_D$$

with an explicitly declared residual $\mathcal{R}_D$.

No later domain introduces an independent selector of primitive evolution.

Proof. The executable selector $\Sigma_{W,\Delta}$ is defined only from the balanced-window executable coordinates, namely $t \bmod W$ and the comparison of the balanced denominator product uv with the floor Δ . Its domain of definition is therefore exactly the executable balanced-window kernel family.

By contrast, the universal lifted-state calculus fixes one parent evolution $\mathcal{E}$ on $\mathcal{X}$. Later branches are introduced by projection and quotient descent. By Theorem 11.2, every later CF branch evolution is derived from the inherited substrate rather than chosen by a new primitive scheduler. Therefore a later branch contributes only a projector Π_D , a visible branch law G_D , and an exactness or residual statement on its declared regime.

Exact state recovery is a separate question. By the solver completeness theorem, unique exact recovery occurs only when the constrained lifted fiber

$$\text{LiftSolve}(\Pi, y, \mathcal{C}, z)$$

is a singleton. This is not selector logic. It is fiber exactness. □

Theorem 11.4 (Projector-tree residual refinement). *Let $\Pi_0 : \mathcal{X} \rightarrow Y_0$ be a parent projector with branch-visible law $G_0 : Y_0 \rightarrow Y_0$, and let*

$$R_0(X) := \Pi_0(\mathcal{E}X) - G_0(\Pi_0 X)$$

be its residual. Suppose a child domain is introduced by a projector

$$\Pi_{1|0} : Y_0 \rightarrow Y_1, \quad \Pi_1 := \Pi_{1|0} \circ \Pi_0,$$

with child-visible law $G_1 : Y_1 \rightarrow Y_1$. Then the child residual is

$$R_1(X) = \Pi_{1|0}(R_0(X)) + \left(\Pi_{1|0}G_0(\Pi_0 X) - G_1(\Pi_1 X) \right).$$

Equivalently,

$$R_1(X) = \Pi_1(\mathcal{E}X) - G_1(\Pi_1 X).$$

Thus a child branch resolves the parent residual by inheriting its projected unresolved part together with the new mismatch created at finer branch resolution.

Proof. Starting from the definition of the child residual,

$$R_1(X) = \Pi_1(\mathcal{E}X) - G_1(\Pi_1 X),$$

and substituting $\Pi_1 = \Pi_{1|0} \circ \Pi_0$, we obtain

$$R_1(X) = \Pi_{1|0}(\Pi_0(\mathcal{E}X)) - G_1(\Pi_1 X).$$

Add and subtract $\Pi_{1|0}G_0(\Pi_0 X)$:

$$R_1(X) = \Pi_{1|0}(\Pi_0(\mathcal{E}X) - G_0(\Pi_0 X)) + (\Pi_{1|0}G_0(\Pi_0 X) - G_1(\Pi_1 X)).$$

The first term is exactly $\Pi_{1|0}(R_0(X))$, which yields the claimed recursion. $\square$

The following table instantiates Theorem 11.2 for each branch. Branches are classified as **exact** (algebraic commutation) or **effective** (commutation on a declared regime with explicit residual type).

T5a. Primitive distinguishability (CF000). Projector: $\Pi_{\text{prim}}(X) = A$ (articulated host class, encoding resolution stage $0D \rightarrow 1D \rightarrow 2D$). Branch evolution: $G_{\text{prim}}(A) = A + 1$ on carry, A otherwise. Status: **exact**. The host-class increment under T is the executable realization of CF000's forced orthogonal re-articulation. No approximation is involved; the quotient diagram commutes algebraically.

T5b. Induced geometry and carrier (CF00). Projector: $\Pi_{\text{carrier}}(X) = (A, q, \vartheta)$ projected onto the smooth carrier $\mathcal{M}$. Branch evolution: G_{carrier} : carrier-level smooth variation inherited from CF00. Status: **effective** on the continuum limit. The phase-calculus state is discrete (tick-valued ϑ); CF00's smooth carrier emerges in the limit $h \rightarrow 0$ of the continuous shadow theorem (Theorem 9.8). Residual: $R = O(h)$, vanishing in the shadow limit.

T5c. QGT and metriplectic split (CF01). Projector: Π_{QGT} : project carrier state onto the quantum geometric tensor decomposition $J \oplus M$. Branch evolution: G_{QGT} : reversible Poisson flow (J -limb) + irreversible metric flow (M -limb) with degeneracy conditions $J \cdot \delta\Sigma = 0$, $M \cdot \delta I = 0$. Status: **effective** on the smooth carrier regime of CF00. The $J \oplus M$ split is a derived consequence of the QGT decomposition on the state family; it inherits the continuum-limit residual from T5b.

T5d. Contact geometry (CF02). Projector: Π_{contact} : project onto the contact manifold (M^{2n+1}, α) . Branch evolution: G_{contact} : contact Hamiltonian flow with Reeb-aligned dissipation. Status: **effective**. Inherits the CF01 effective regime. The contact structure is the odd-dimensional geometric reading of the metriplectic split.

T5e. Hierarchical interfaces (CF03). Projector: Π_{hier} : project onto boundary-concentration functional and hierarchy depth $N(L)$. Branch evolution: G_{hier} : logarithmic scaling $N(L) \leq \lceil \log_2(L/\ell_0) \rceil + 1$. Status: **exact** on the executable balanced-window branch. The truncated

realized depth $\delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}$ is the executable realization of the A8/CF03 logarithmic hierarchy law, sourced through the Farey remainder recursion. The refinement depth $d_A(\Delta) = O(\log \Delta)$ is the compressed bridge form of A8.

T5f. Telegraph-Fisher causality (CF04). Projector: Π_{cone} : project onto transport-cone structure (finite-speed propagation). Branch evolution: G_{cone} : telegraph-Fisher dynamics with $c_{\text{eff}} = c_0 \exp(-\frac{1}{2}\beta D)$. Status: **effective** on the continuum transport limit. The one-tick phase advance $\pi/2$ per step is the discrete precursor; the finite-speed cone emerges in the continuum limit with residual $O(h)$.

T5g. Integrability closure (CF05). Projector: Π_{int} : project onto the space of first integrals of a declared finite-dimensional truncation. Branch evolution: G_{int} : Darboux / Prolle-Singer search certifying no hidden conserved quantities beyond I, Σ . Status: **effective** on the declared truncation. The lifted conservation criterion (Theorem 8.9, $\text{Diff } Q = 0$) is exact; the CF05 integrability search operates on a finite-dimensional projection thereof.

T5h. Information geometry (CF06). Projector: Π_{info} : project onto the statistical manifold with Fisher metric g_{ij}^F . Branch evolution: G_{info} : natural gradient flow $\dot{\theta}^i = -g_F^{ij} \partial_j \Sigma$ as M-limb dissipation. Status: **effective** on the coarse-grained statistical-manifold regime. The Fisher metric is the intrinsic distinguishability metric on the coarse-grained probability family; it inherits the M-limb structure from CF01.

T5i. Measurement and decoherence (CF07). Projector: Π_{meas} : project onto pointer-basis reduced state ρ_S . Branch evolution: G_{meas} : CPTP dephasing channel $\mathcal{E}(\rho) = \sum_i \Pi_i \rho \Pi_i$ as $J \rightarrow M$ projection. Status: **effective** on the decoherence regime (einselection residual below tolerance, decoherence time τ_{dec} within stated window). The measurement channel is the coarse-grained reading of the lifted evolution restricted to pointer-stable subspaces.

T5j. Spinor emergence (CF08). Projector: Π_{spin} : project onto the low-energy domain-wall fermion sector. Branch evolution: G_{spin} : domain-wall fermion propagation with Nielsen-Ninomiya compliance. Status: **effective** on the low-energy / adiabatic window. The spinor bundle is a derived low-energy sector of the lattice-level domain-wall construction.

T5k. Gauge emergence (CF09). Projector: Π_{gauge} : project onto Berry connection $A_\mu = -i\langle\psi|\partial_\mu\psi\rangle$. Branch evolution: G_{gauge} : Maxwell dynamics from the leading $\int F_{\mu\nu} F^{\mu\nu}$ effective action on the declared IR window (gates G1–G6). Status: **effective** on the IR / derivative-expansion window. The Berry connection is a derived geometric object on the low-energy state family; its curvature is the emergent field strength. Higher-derivative corrections are subleading on the declared window.

T5l. Hydrodynamics (CF10). Projector: Π_{hydro} : project onto coarse-grained density/momentum fields. Branch evolution: G_{hydro} : Navier-Stokes equations via lattice Boltzmann continuum limit. Status: **effective** on the hydrodynamic scaling regime. Residual: lattice-scale corrections.

T5m. Dark sector (CF11). Projector: Π_{dark} : project onto cosmological background with dark-sector split. Branch evolution: G_{dark} : FRW-type evolution with metriplectic dark energy/matter sourced from the $J \oplus M$ split (CF01), the hierarchical boundary law (CF03), and finite-speed transport (CF04). Status: **effective** on the cosmological / FRW regime. Residual: sub-horizon corrections and back-reaction (closed in CF12).

T5n. Gravity (CF12). Projector: Π_{grav} : project onto Lorentzian metric $g_{\mu\nu}$ with burden source $T_{\mu\nu}^{\text{inv}}$. Branch evolution: G_{grav} : Einstein-form effective field equation with saturation corrections $\Delta_{\mu\nu}^{\text{sat}}$. Status: **effective** on the weak-field / low-curvature regime. The Lorentzian metric inherits its null cone from CF04, its source from the burden functional (CF00/CF01), and its saturation sector from CF000/CF03. The Einstein-form closure is the unique local second-order divergence-compatible closure in four dimensions. Residual: near-saturation corrections parametrized by $\Delta_{\mu\nu}^{\text{sat}}$.

T5o. π , transcendence, and chirality (CF13). Projector: Π_{chiral} : project onto chiral sector of domain-wall spectrum. Branch evolution: G_{chiral} : chiral transport with Nielsen–Ninomiya obstruction. Status: **effective** on the low-energy chiral window. The status of π as a visible-carrier witness (Theorem 6.6) is exact; the chiral domain-wall realization inherits the CF08 effective regime.

T5p. Stationary action (CF14). Projector: Π_{action} : project onto temporal path with articulation cost L . Branch evolution: G_{action} : Euler–Lagrange equation as zero first-variation condition. Status: **effective** in the continuum shadow limit. The exact lifted accumulation (Definition 8.4) is the discrete precursor; the stationary-action principle emerges through the continuous shadow theorem (Theorem 9.8) and is the temporal effective expression of the primitive non-discharge law (A(−1)/CF000).

T5q. Noether conservation (CF15). Projector: Π_{Noether} : project onto zero-cost symmetry directions. Branch evolution: G_{Noether} : conserved currents $\partial_\mu j^{\mu,a} = 0$ (abelian) or $D_\mu j^{\mu,a} = 0$ (non-abelian). Status: **effective** on the CF14 continuum regime. The exact conservation criterion (Theorem 8.9, $\text{Diff } Q = 0$) is the discrete ancestor; Noether’s theorem is its continuum-shadow expression in zero-cost directions. The J -limb exactly conserves Noether charges; the M -limb breaks them with drift proportional to entropy production rate.

T5r. Electroweak splitting (CF16). Projector: Π_{EW} : project onto $\text{SU}(2)_L \times \text{U}(1)_Y$ fiber structure. Branch evolution: G_{EW} : electroweak dynamics with chiral coupling and gauge-scalar breaking. Status: **effective** on the electroweak scale window. The fiber splitting is the non-abelian extension of the CF09 Berry connection construction, specialized to the left-chiral sector.

T5s. Non-abelian confinement (CF17). Projector: Π_{conf} : project onto color flux-tube sector. Branch evolution: G_{conf} : Wilson loop area law, string breaking, deconfinement transition. Status: **effective** on the strong-coupling / large-distance regime. Confinement is the non-abelian saturation phenomenon; the Wilson loop area law is the strong-coupling descendant of CF09 gauge emergence.

T5t. Renormalization and running couplings (CF18). Projector: Π_{RG} : project onto scale-dependent effective couplings. Branch evolution: G_{RG} : beta functions from scale re-articulation with

hidden-return corrections. Status: **effective** on the perturbative / EFT regime. Scale re-articulation is the mechanism by which the primitive invariant's cost is redistributed across scales; running couplings are its effective-field-theoretic expression.

11.3 Worked Example: Native Packet-Collapse Descent and Bring Quintic Certification

We record the shortest concrete demonstration of the method on the native quintic surface.

Start from the exact Red quotient

$$\Pi_{\text{Red}} \circ B = G_{\text{Red}} \circ \Pi_{\text{Red}}, \quad \Pi_{\text{Red}}(\hat{\Xi}) = q = (u, v), \quad G_{\text{Red}}(u, v) = \text{sort}(v, u + v).$$

The scalar descendants are not imported from outside. They are read from the native corridor pair:

$$\rho(u, v) = \log(uv), \quad \mathcal{R}(u, v) = E\left(\frac{2\pi}{uv}\right),$$

with

$$E(t) = \frac{\log(q; q)_{\infty} + \pi^2/(6t) - \frac{1}{2} \log(2\pi/t)}{t}, \quad q = e^{-t}.$$

Now descend along the balanced corridor to the canonical anchor

$$(u, v) = (55, 89).$$

At this anchor one has

$$uv = 4895, \quad t = \frac{2\pi}{4895}, \quad c^* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right].$$

The native packet-collapse descendant therefore evaluates to

$$\mathcal{R}(55, 89) = E\left(\frac{2\pi}{4895}\right).$$

By the arithmetic-completion bridge, this locks to the exact edge residual

$$E\left(\frac{2\pi}{4895}\right) \rightarrow \frac{1}{24}.$$

So the canonical anchor is not merely a corridor waypoint. It is the first place where the native Red descendant, the balanced germ, and the packet-collapse law coincide on one explicit finite witness.

This already shows the operative order of the method. The corridor evolution is carried in retained lifted state first. The scalar descendant is read only after the retained pair has been transported to the anchor. Nothing in this step requires scalar radical discharge.

The practical quintic witness then uses the same retained-first order on the Bring quintic

$$x^5 - x + 1 = 0.$$

The shipped certifier operates on the retained branch-complete state and returns all five projected roots only after certification has been carried through the lifted transport. In the release bundle, all five Bring roots certify at depth 21, with half-width below 10^{-8} , and with projected polynomial residuals near machine precision. This is the immediate executable form of the paper's claim: retained transport first, projected root readout second.

Accordingly, the worked example has two layers. The first layer is the native packet-collapse descent

$$(55, 89) \mapsto E\left(\frac{2\pi}{4895}\right) = \frac{1}{24},$$

which demonstrates that the Red quotient is genuinely internal to the balanced corridor. The second layer is the Bring certification witness, which demonstrates that the same lifted-state transport discipline recovers concrete quintic roots and verifies them against the original polynomial.

11.4 Cross-domain quotient descent theorems

The T5 inventory declares the relevant projectors and branch evolutions. The purpose of the present section is stronger: it writes the explicit quotient-descent statements showing that the known branch equations of physics are the shadow of the lifted evolution on the stated projections. Exact branches descend by algebraic commutation; effective branches descend by Theorem 11.2 together with the continuous shadow theorem.

U1. Conservation laws (CF15)

Theorem 11.5 (Quotient descent of Noether conservation). *Let $\Pi_{\text{Noether}} : \mathcal{X} \rightarrow Y_{\text{Noether}}$ be the T5q projector onto the zero-cost symmetry sector of CF15, and let Q be any transported lifted charge satisfying*

$$\text{Diff}(Q \circ \Pi_{\text{Noether}}) = 0.$$

Then on the CF14/CF15 continuum-shadow regime,

$$\lim_{h \rightarrow 0} \text{Rate}_{T_h}(Q \circ \Pi_{\text{Noether}}) = 0,$$

and, under the CF15 identification of the projected transport with Noether-current transport,

$$\partial_\mu j^{\mu,a} = 0$$

for abelian currents and

$$D_\mu j^{\mu,a} = 0$$

for non-abelian currents.

Proof. The exact lifted ancestor is the conservation criterion $\text{Diff } Q = 0$. By Theorem 11.2, the Noether branch is the quotient of the lifted evolution on the declared zero-cost symmetry projector. By the continuous shadow theorem, exact discrete conservation descends to vanishing continuous rate. CF15 identifies the projected continuous transport law with Noether-current conservation, yielding $\partial_\mu j^{\mu,a} = 0$ and its gauge-covariant non-abelian form $D_\mu j^{\mu,a} = 0$. $\square$

U2. Stationary action and Euler–Lagrange descent (CF14)

Theorem 11.6 (Quotient descent of stationary action). *Let Π_{action} be the $T5p$ projector onto the temporal path together with articulation-cost density L . Then the exact lifted accumulation*

$$\text{Acc}_T[L; a, b](X_a) = \sum_{k=a}^{b-1} L(X_k) \text{Diff } T(X_k)$$

descends, on the CF14 continuum-shadow regime, to the action functional

$$S[q] = \int_{t_a}^{t_b} L(q, \dot{q}, t) dt.$$

Moreover, the lifted stationarity condition descends to

$$\delta S = 0,$$

and hence to the Euler–Lagrange equation

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q}.$$

Proof. Definition 8.4 supplies the exact lifted accumulation. The continuous shadow theorem sends the Riemann-sum form of Acc_T to the integral action in the limit $h \rightarrow 0$. CF14 proves that the temporal effective invariant is precisely absence of first-order excess articulation cost on endpoint-fixed paths; this is the branch evolution G_{action} on the action projector. Therefore $\Pi_{\text{action}} \circ \mathcal{E} = G_{\text{action}} \circ \Pi_{\text{action}}$ on the declared regime, and the descended branch equation is the Euler–Lagrange law. $\square$

U3. Gravity and Einstein-form descent (CF12)

Theorem 11.7 (Quotient descent of gravity). *Let $\Pi_{\text{grav}} : \mathcal{X} \rightarrow Y_{\text{grav}}$ be the $T5n$ projector onto the Lorentzian metric sector and burden source tensor. On the CF12 weak-field / low-curvature regime,*

$$\Pi_{\text{grav}} \circ \mathcal{E} = G_{\text{grav}} \circ \Pi_{\text{grav}} + R_{\text{grav}}, \quad R_{\text{grav}} \rightarrow 0$$

on the stated shadow window. Under the CF12 identification of the projected branch evolution with burden-gradient geometry, the descended field equation is

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu}^{\text{inv}} + \Delta_{\mu\nu}^{\text{sat}}.$$

Proof. The gravity projector is already declared in T5n. CF12 derives the Lorentzian metric from the finite-speed cone inherited through CF04 and derives the Einstein-form closure as the unique local second-order divergence-compatible response to nonuniform burden. By Theorem 11.2, this branch evolution is the quotient of the inherited substrate on the CF12 declared regime. The residual is the usual shadow / low-curvature error outside that window; on the declared regime it vanishes and the descended equation is the Einstein-form law with saturation correction sector. $\square$

U4. Electromagnetism and Maxwell descent (CF09)

Theorem 11.8 (Quotient descent of Maxwell dynamics). *Let $\Pi_{\text{gauge}} : \mathcal{X} \rightarrow Y_{\text{gauge}}$ be the T5k projector onto the Berry-connection sector,*

$$A_\mu = -i\langle\psi | \partial_\mu \psi\rangle, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

On the CF09 infrared / derivative-expansion regime,

$$\Pi_{\text{gauge}} \circ \mathcal{E} = G_{\text{gauge}} \circ \Pi_{\text{gauge}} + R_{\text{gauge}}, \quad R_{\text{gauge}} \rightarrow 0$$

as the declared window is approached. The descended field equation is Maxwell form:

$$\partial_\mu F^{\mu\nu} = J^\nu,$$

with leading effective action

$$\int F_{\mu\nu} F^{\mu\nu}.$$

Proof. CF09 derives the emergent gauge potential as the Berry connection on the low-energy spinor bundle imported from CF08 and derives the leading gauge-invariant Maxwell operator on the infrared window. T5k declares exactly this projector and branch evolution. By Theorem 11.2, the quotient diagram commutes on the stated regime, while the residual collects higher-derivative corrections outside it. Thus the projected shadow of lifted evolution is the Maxwell branch equation. $\square$

U5. Measurement channel and Born rule descent (CF07)

Theorem 11.9 (Quotient descent of measurement and Born weights). *Let $\Pi_{\text{meas}} : \mathcal{X} \rightarrow Y_{\text{meas}}$ be the T5i projector onto the pointer-basis reduced state ρ_S . On the CF07 decoherence / einselection regime,*

$$\Pi_{\text{meas}} \circ \mathcal{E} = G_{\text{meas}} \circ \Pi_{\text{meas}} + R_{\text{meas}}, \quad G_{\text{meas}}(\rho) = \sum_i \Pi_i \rho \Pi_i, \quad R_{\text{meas}} \rightarrow 0.$$

The descended measurement law is the CPTP dephasing channel above, and the corresponding outcome weights obey

$$P(i \mid \psi) = |\langle i \mid \psi \rangle|^2.$$

Proof. CF07 derives decoherence as a channel on the reduced state and proves that the Born-rule weight is the unique phase-invariant, compositional, normalized choice compatible with the measurement axioms used there. T5i already declares the pointer-basis projector and the dephasing branch evolution. Theorem 11.2 gives commutation on the stated decoherence regime, and the continuous-shadow interpretation of the $J \rightarrow M$ projection yields the irreversible measurement channel. The Born probability formula is then the branch-visible weight attached to that projected channel. $\square$

U6. Thermodynamics and entropy production descent (CF01, CF06)

Theorem 11.10 (Quotient descent of the second law). *Let Π_{QGT} and Π_{info} be the T5c/T5h projectors onto the metriplectic and information-geometric sectors. If Σ is the lifted entropy view with*

$$\text{Diff } \Sigma \geq 0,$$

then on the CF01/CF06 coarse-grained statistical regime,

$$\lim_{h \rightarrow 0} \text{Rate}_{T_h}(\Sigma \circ \Pi_{\text{info}}) \geq 0,$$

and the descended thermodynamic law is

$$\frac{dS}{dt} \geq 0.$$

Equivalently, the M-limb descends to the information-geometric natural-gradient flow

$$\dot{\theta}^i = -g_F^{ij} \partial_j \Sigma.$$

Proof. CF01 derives the exact $J \oplus M$ split with M -limb entropy nondecrease, and CF06 identifies the metric limb with Fisher / Ruppeiner information geometry on the declared statistical manifold. The lifted ancestor is the exact production criterion $\text{Diff } \Sigma \geq 0$. By Theorem 11.2, the information-geometric branch is the quotient of the metriplectic substrate on its declared coarse-grained regime. Passing to the continuous shadow gives $dS/dt \geq 0$ and the natural-gradient form of dissipative evolution. $\square$

U7. Finite-speed propagation and Lorentzian cone descent (CF04, CF12)

Theorem 11.11 (Quotient descent of finite-speed propagation). *Let Π_{cone} be the T5f projector onto the transport-cone structure. On the CF04 transport regime,*

$$\Pi_{\text{cone}} \circ \mathcal{E} = G_{\text{cone}} \circ \Pi_{\text{cone}} + R_{\text{cone}}, \quad R_{\text{cone}} = O(h),$$

and the descended continuous branch law is the telegraph equation

$$\tau \partial_{tt} u + \partial_t u = D \Delta u, \quad c = \sqrt{D/\tau}.$$

The same descended cone, when read through the gravity projector of CF12, gives the Lorentzian null condition

$$g_{\mu\nu} dx^\mu dx^\nu = 0.$$

Proof. CF04 derives finite-speed propagation from the Cattaneo–Vernotte closure and identifies the characteristic transport cone of the telegraph equation. T5f declares that branch. The lifted finite-step ancestor is the bounded one-tick propagation of the executable evolution; by the continuous shadow theorem this becomes the telegraph equation on the declared regime. CF12 then inherits that same propagation-limiting cone to fix the Lorentzian sign structure, so the projected cone becomes the null cone of the metric. $\square$

U8. Spinor structure and Dirac descent (CF08)

Theorem 11.12 (Quotient descent of the spinor branch). *Let Π_{spin} be the T5j projector onto the low-energy domain-wall fermion sector. On the CF08 low-energy / adiabatic regime,*

$$\Pi_{\text{spin}} \circ \mathcal{E} = G_{\text{spin}} \circ \Pi_{\text{spin}} + R_{\text{spin}}, \quad R_{\text{spin}} \rightarrow 0.$$

At the lattice-visible stage the descended branch obeys the Ginsparg–Wilson relation

$$\{D, \gamma_5\} = aD\gamma_5D,$$

and in the continuum shadow limit it yields the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

Proof. CF08 constructs the sign-changing domain-wall mass profile and proves the existence of exponentially localized chiral zero modes whose low-energy effective operator satisfies the Ginsparg–Wilson relation. T5j declares precisely this projector and effective regime. Theorem 11.2 gives the branch commutation on that regime; the continuous shadow theorem then identifies the continuum limit of the low-energy wall mode with the Dirac equation. $\square$

U9. Electroweak descent (CF16)

Theorem 11.13 (Quotient descent of the electroweak branch). *Let Π_{EW} be the T5r projector onto the $\text{SU}(2)_L \times \text{U}(1)_Y$ fiber structure. On the CF16 electroweak regime,*

$$\Pi_{\text{EW}} \circ \mathcal{E} = G_{\text{EW}} \circ \Pi_{\text{EW}} + R_{\text{EW}}, \quad R_{\text{EW}} \rightarrow 0.$$

The descended branch law is electroweak gauge dynamics with covariant derivative

$$D_\mu = \partial_\mu - ig W_\mu^a \frac{\tau^a}{2} - ig' Y B_\mu,$$

neutral gauge mixing into one massless and one massive direction, and tree-level mass relation

$$m_W = m_Z \cos \theta_{EW}.$$

Proof. CF16 derives electroweak hosting from the already-earned abelian phase direction, the minimal non-abelian internal rotation sector, the chiral boundary-hosted matter family, and the tachyonic order parameter. T5r records this branch as the non-abelian quotient descendant of CF09 specialized to the left-chiral sector. The derivation-chain exactness theorem therefore gives commutation on the electroweak regime, and the descended visible branch equations are exactly the electroweak covariant derivative, neutral mixing, and W/Z mass law derived in CF16. $\square$

U10. Confinement and Wilson-area descent (CF17)

Theorem 11.14 (Quotient descent of non-abelian confinement). *Let Π_{conf} be the T5s projector onto the color flux-tube sector. On the CF17 strong-coupling / large-distance regime,*

$$\Pi_{\text{conf}} \circ \mathcal{E} = G_{\text{conf}} \circ \Pi_{\text{conf}} + R_{\text{conf}}, \quad R_{\text{conf}} \rightarrow 0.$$

The descended confinement law is

$$F_{\text{conf}}(R) = \sigma_{\text{conf}} R + o(R),$$

and equivalently the Wilson-loop area law

$$\langle W(C) \rangle \sim \exp(-\sigma \text{Area}(C)).$$

Proof. CF17 derives confinement as non-abelian saturation of the color sector inherited from CF09/CF16. T5s declares the corresponding flux-tube projector and confinement branch evolution. By Theorem 11.2, this branch is the quotient descendant of the inherited non-abelian substrate on the stated strong-coupling regime. The projected large-distance law is the linear confinement observable, which is equivalent to the Wilson-loop area law on the same regime. $\square$

U11. Running couplings and beta-function descent (CF18)

Theorem 11.15 (Quotient descent of running couplings). *Let Π_{RG} be the T5t projector onto the scale-dependent effective couplings. On the CF18 EFT / perturbative regime,*

$$\Pi_{\text{RG}} \circ \mathcal{E} = G_{\text{RG}} \circ \Pi_{\text{RG}} + R_{\text{RG}}, \quad R_{\text{RG}} \rightarrow 0.$$

The descended renormalization-group law is

$$\beta_i(\mu) = \mu \frac{dg_i}{d\mu},$$

with hidden-return corrections encoded in the effective action.

Proof. CF18 derives reduced description, effective articulation-cost functional, hidden-return corrections, running coefficients, and beta functions as scale re-articulation of the primitive invariant. T5t declares the scale-dependent coupling projector and the RG branch evolution. By Theorem 11.2, the branch is the quotient descendant of the lifted evolution on the declared EFT regime. The projected scale-flow equation is therefore exactly the beta-function law $\beta_i(\mu) = \mu dg_i/d\mu$. $\square$

U12. Hierarchical scaling descent (CF03)

Theorem 11.16 (Quotient descent of hierarchical scaling). *Let Π_{hier} be the T5e projector onto the hierarchy / boundary-concentration sector. On the executable balanced branch,*

$$\delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\} = O(\log \Delta).$$

Under the CF03 continuum identification of the visible resolution ratio with L/ℓ_0 , the descended hierarchy law is

$$N(L) \leq \left\lceil \log_2 \left(\frac{L}{\ell_0} \right) \right\rceil + 1,$$

with realizable logarithmic hierarchy depth $N(L) = \Theta(\log(L/\ell_0))$.

Proof. T5e already identifies the executable truncated realized depth as the discrete realization of the A8 / CF03 hierarchy law, sourced through the Farey balanced-refinement bridge. The refinement count $d_A(\Delta)$ grows logarithmically with the floor parameter on the balanced corridor, and the macro-horizon clamp preserves that order. CF03 derives the continuum logarithmic bound on active hierarchy depth between macroscopic scale L and cutoff ℓ_0 . Thus the discrete corridor depth is the quotient descendant of the same hierarchy law, and the continuum branch equation is the stated logarithmic scaling relation. $\square$

11.5 Scope statement for domain descent

The branch-translation interface is therefore not a loose metaphor. It is a hard exactness filter:

- (i) choose the declared branch-visible coordinates,
- (ii) write the candidate descendant law,
- (iii) test the commuting quotient relation,
- (iv) transport conserved or produced scalar views by composition,

- (v) state the residual and scope whenever exactness fails.

Chapter 12

Closure Certificate

Assumption ledger

- (A1) The primitive visible roll is exactly $z(\theta) = ie^{i\theta}$.
- (A2) The balanced lifted object law and the balanced refinement law are already earned and are used extensionally here.
- (A3) The next-host map $\mathcal{N}$ is lawful wherever host lift is applied.
- (A4) The balanced-window family supplies the explicit computational return witness and the balanced identities proved here; universal exact return is forced by non-discharge, the universal witness pattern, and finite stopping depth under any finite floor.
- (A5) The selector-complete executable theorems hold for all admissible regime pairs (W, Δ) with $W \geq 1$ and $\Delta \geq 1$ (Theorem 7.19). Numerical verification has been performed for $(W, \Delta) = (64, 4096)$.
- (A6) The canonical launch family $A = 0, 1, 2$ with anchor $(55, 89)$ and launch height $1/4895$ is the numerically verified instance of the algebraically universal balanced-corridor mock-theta mechanism.
- (A7) A shadow completion coordinate is attached only after a shadow form g is fixed.

Limits of applicability

- (L1) The selector-complete macro-calculus is regime-universal for all admissible executable balanced-window regime pairs (W, Δ) . The canonical block word $R^{\delta_A} S^{W-1-\delta_A} T$ with $\delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}$ is the unique reachable trajectory. Numerical verification has been performed for $(64, 4096)$.
- (L2) Exact return is witnessed computationally on the balanced-window family and proved universal by non-discharge (CF000), the universal witness pattern (H000), and finite stopping depth (Farey bridge). CF19 data confirms convergence across 16 seed families.

- (L3) The constructive VDM-to-mock-theta identification is algebraically universal on the balanced corridor. The canonical launch family $A = 0, 1, 2$ is the numerically verified instance. The analytic contour-equivalence in the sense of Zwegers’ thesis is not attempted here.
- (L4) Later branch formalisms are classified as **exact** (algebraic commutation with the lifted evolution) or **effective** (commutation on a declared regime with explicit residual type). Per-branch status is inventoried in the T5 interface (Section 11.2). Each effective branch inherits exactness through derivation-chain commutation (Theorem 11.2) on the regime where its stated approximations hold.
- (L5) Projector-preserved real-analyticity and Liouvillian closure are claimed only for the continuous-shadow exp-log class $\mathcal{C}_{\text{sh}}$ on natural domains, not for arbitrary external function classes or arbitrary projectors outside the stated quotient discipline.
- (L6) The native π streaming, bank-indexed access, and Bring-certifier timing values are operational witness values of the bundled implementations. They certify existence and practical strength of the selector-closed engine on those branch surfaces; they are not machine-independent asymptotic performance bounds.

Inheritance manifest

This merged manuscript directly incorporates theorem-bearing content from the primitive-roll / lifted-object stack, the selector-closed executable package, the connection-closure note, and the CF19 full lifted-object branch. In particular it carries forward:

1. the primitive roll of i and the half-turn constant π ,
2. the full lifted object $\hat{\Xi}$ and the balanced-window forward law,
3. the exact connection-closure identities $\mathbf{Q} = F_{\text{same}}$, $\mathbf{B} = F_{\text{ref}}$, $\mathbf{L} = F_{\perp}$,
4. the selector-closed executable macro-grammar $\{\mathbf{R}, \mathbf{S}, \mathbf{T}\}$,
5. the arithmetic-completion bridge, nonladder transport, and canonical launch-family branch,
6. the VDM-first mock-theta derivation and its first coefficient data.

12.1 Inheritance Manifest

The present formalism inherits the following primitive and structural results:

- **Universal Orthogonality Invariant: The Generative Nature of i** Zenodo DOI: 10.5281/zenodo.19211725 Primitive roll definition and unique origin law.

- **CF000: Primitive Distinguishability and Orthogonal Articulation** Zenodo DOI: 10.5281/zenodo.19380587 Forced admissible origin, non-discharge, and orthogonal re-articulation.
- **Farey Remainder Recursion and Orthogonal Re-Articulation** Zenodo DOI: 10.5281/zenodo.19210954 Exact denominator-pair update and balanced refinement law $B(u, v) = \text{sort}(v, u + v)$.

All downstream CF results used in this manuscript are either:

(i) re-proved explicitly, or (ii) inherited through CF19 integration without reopening their theorem burden.

Closed in this manuscript

Closed burden	Resolution in this manuscript
Primitive source witness	The roll $z(\theta) = \text{ie}^{i\theta}$
Half-turn constant	Continuous completion with primitive π
Exact state	Full lifted object $\hat{\Xi}$
Finite operator core	Primitive operator alphabet $\mathcal{U} = \{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}$
Connection closure	Individual proofs that $\mathbf{Q} = F_{\text{same}}, \mathbf{B} = F_{\text{ref}}, \mathbf{L} = F_{\perp}$
Finite evolution	Exact operator-word evolution
Selector burden on executable branch	Macro grammar $\{\mathbf{R}, \mathbf{S}, \mathbf{T}\}$ and selector law $\Sigma_{W, \Delta}$
Executable regime witness	Canonical block word, current-regime corollary, and exact 256-step verification statement
Balanced exact laws	Fibonacci branch, germ law, recurrence criterion, commutation identities
Exact return witness	Witnessed exact return to lifted origin on the balanced-window family
Visible versus full closure	Theorem that π is a visible-carrier witness and not full-object closure
Arithmetic-completion bridge	Full derivation to $(55, 89)$, $r_* = 1/4895$, $E(t) \rightarrow 1/24$, $G(t) \rightarrow 1/12$
Nonladder host transport	Survivor-class invariance across host change with tested seeds and host period 16
Completion branch	Shadow Calculus as downstream completion/residual layer
Launched-family branch	VDM-first mock-theta derivation, first coefficients, constructive identification

Universal derivative/integral layer	Exact lifted difference, rate, accumulation, and solver criteria
Universality criterion	Exact quotient theorem $\Pi \circ \mathcal{E} = G \circ \Pi$
Quotient instances	Visible quarter closure, Fibonacci branch, raw-engine phase quotient
Classical calculus recovery	Continuous shadow theorem
Domain descent	Universal applicability theorem and branch-translation interface items T1–T5
Primitive operation descent chain	Explicit theorem-grade canonical spine from survivor mark to continuous shadow
Operational native witness surface	Certified native π streaming, bank-indexed access, and depth-21 Bring quintic certification as direct execution witnesses of the selector-closed lifted engine
Projector lower bound	State-complete projector simulation lower bound and the corollary that branch readout is not primitive state
Projector-preserved invariants	Real-analyticity on natural domains and Liouvillian exp-log closure of the continuous-shadow class $\mathcal{C}_{\text{sh}}$
Continuous-shadow worked example	Lifted rotation readout yielding $(\sin x, \cos x)$
Branch-memory register	Generic retained branch-complete state

$$\widehat{\Xi}_{\mathcal{B}} = (A, q, \theta, \kappa, c, s, h),$$

sheet transport law

$$\Pi_s \circ E_g = \rho(g) \circ \Pi_s,$$

history transport law

$$\Pi_h \circ E_g = R_g \circ \Pi_h,$$

branch-memory quotient law

$$\Pi_{\mathcal{B}} \circ E_g = G_g \circ \Pi_{\mathcal{B}},$$

and projection-loss theorem for stripped readouts that omit s or h .

Decisive falsifiers

The manuscript fails if any of the following occur:

- (F1) the primitive operators fail to preserve the defining lifted-state relations;
- (F2) visible witness data become state-complete on the full lifted state;
- (F3) balanced refinement from $(1, 1)$ fails the Fibonacci corridor law;
- (F4) the completion germ fails to be determined by (q, θ) on the balanced family;
- (F5) the selector $\Sigma_{W, \Delta}$ fails to reproduce the verified executable macro schedule on the fixed regime;
- (F6) a purported exact branch law fails the quotient commutation test;
- (F7) a purported exact solver is not the singleton solution of a constrained lifted fiber.

Claims established here

- (C1) exact lifted state space and visible witness,
- (C2) coordinate projectors and germ determinacy,
- (C3) primitive operators $\mathbf{Q}, \mathbf{B}, \mathbf{L}$ and their exact inherited identities,
- (C4) operator-relative observable difference and finite accumulation,
- (C5) exact operator-word evolution,
- (C6) balanced Fibonacci corridor, balanced germ law, and first commuting identities,
- (C7) witnessed exact return to lifted origin on the balanced-window family,
- (C8) visible-carrier witness status of π ,
- (C9) canonical balanced floor anchor and arithmetic-completion bridge,
- (C10) nonladder host transport invariance,
- (C11) selector-closed executable macro-calculus $\{\mathbf{R}, \mathbf{S}, \mathbf{T}\}$,
- (C12) state-only selector theorem, canonical block word, and exact 256-step witness,
- (C13) universal lifted-state calculus and exact quotient theorem,
- (C14) three proved quotient instances,
- (C15) continuous shadow theorem,

- (C16) VDM-first mock-theta derivation, first VDM-generated coefficients, and constructive VDM-to-mock-theta identification,
- (C17) universal applicability theorem and branch-translation interface T1–T5,
- (C18) regime-universal selector law with truncated realized depth $\delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}$ for all admissible (W, Δ) ,
- (C19) normal form completeness: canonical block words are the only reachable words on the executable branch,
- (C20) derivation-chain exactness theorem and per-branch projector inventory (T5a–T5t) classifying all CF branches as exact or effective on declared regimes,
- (C21) universal exact return forced by non-discharge: the primitive invariant cannot stop being borne (CF000), orthogonal re-articulation is universal (H000 Universal Axiom), finite stopping depth holds under any finite floor (Farey bridge), and CF19 data confirms convergence across 16 seed families and adversarial testing at widths 16 and 32,
- (C22) universal mock-theta mechanism on the balanced corridor: the algebraic derivation (centered germ, two-edge transport kernel, shell recursion) depends only on the balanced corridor structure $B(u, v) = \text{sort}(v, u + v)$ and applies to any launch height $1/(uv)$, not only the canonical family $A = 0, 1, 2$,
- (C23) automatic derivation-chain exactness within declared regimes: every CF branch evolution is derived from the inherited substrate (Theorem 11.2), and every later axiom is an effective expression of the primitive invariant within an admitted domain (H000 Theorem 7.3), so commutation is a structural theorem on each CF’s declared regime.
- (C24) twelve explicit cross-domain quotient-descent theorems recovering the known branch equations of physics — Noether conservation, stationary action, Einstein-form gravity, Maxwell dynamics, the measurement/Born channel, the second law, the finite-speed/Lorentzian cone, Dirac spinors, electroweak structure, confinement, renormalization-group flow, and logarithmic hierarchy scaling — as shadows of the lifted evolution.
- (C25) primitive operation descent chain from survivor mark to continuous shadow,
- (C26) state-complete projector simulation lower bound and the corollary that branch readout is not primitive state,
- (C27) projector-preserved real-analyticity on natural domains and projector-preserved Liouvillian exp-log closure of the continuous-shadow class $\mathcal{C}_{\text{sh}}$,
- (C28) operational native witness surface: 1,000,000 native π digits in 1.65086778 s, 1024-digit bank-indexed access at 5.39419×10^{-7} s/query, and depth-21 Bring quintic certification with final half-width $6.18979215099812 \times 10^{-9}$,

(C29) worked continuous-shadow descent from lifted rotation to $(\sin x, \cos x)$.

Claims not established here

- (N1) a full contour-integral proof of mock-theta modularity beyond the constructive VDM-to-mock-theta identification (the algebraic identification is complete; the analytic contour-equivalence in the sense of Zweegers' thesis is not attempted here),
- (N2) extension of the executable macro-calculus to non-balanced-window kernel families (e.g. non-Fibonacci corridor arithmetic operators).

Red-team leak checklist

The document fails its own discipline if any of the following leak in:

- (R1) selector logic is used outside the executable balanced-window family as though already closed for non-balanced-window regimes,
- (R2) visible recurrence is treated as though it were full lifted-object return,
- (R3) π -position events are used as though they alone closed the full carried state,
- (R4) shadow/residual formulas are used as primitive sources rather than downstream completion data,
- (R5) raw state subtraction is treated as the primitive derivative notion,
- (R6) branch exactness is claimed without stating the derivation regime and residual type,
- (R7) return claims are made on non-balanced-window families without extending the present formalism to those families,
- (R8) effective-branch commutation is treated as exact outside the declared approximation regime,
- (R9) unification claims stronger than the current exact quotient translations support are asserted.
- (R10) sheet equality is treated as branch-history equality;
- (R11) visible equality is treated as monodromy faithfulness;
- (R12) a branch law is claimed without recording the group presentation, sheet action, and retained history update.

Chapter 13

Conclusion

Phase Calculus is the whole system. Its primitive center is the pair

$$z(\theta) = \text{ie}^{i\theta}, \quad \hat{\Xi} = (A, q, \theta, \kappa, c).$$

Its finite exact operator core is

$$\mathcal{U} = \{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}.$$

Its regime-universal executable closure is the selector-complete macro-grammar

$$\{\mathbf{R}, \mathbf{S}, \mathbf{T}\}, \quad \Sigma_{W,\Delta}(X) \in \{\mathbf{R}, \mathbf{S}, \mathbf{T}\},$$

valid for all admissible (W, Δ) , with unique reachable block words $\mathbf{R}^{\delta_A} \mathbf{S}^{W-1-\delta_A} \mathbf{T}$ and truncated realized depth $\delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}$. Its universal calculus layer is the triple

$$(\mathcal{X}, \mathcal{E}, \mathcal{V}),$$

with exact difference, rate, accumulation, fiber-exact solution, symmetry transport, and quotient descent. Every branch of the Complete Formalism series (CF000–CF18) descends from this substrate through a declared projector and derivation-chain commutation, classified as exact or effective on its declared regime. Later domains do not add selectors; they add projectors and branch-visible laws descending from the one lifted evolution. The manuscript now states the corresponding quotient-descent theorems explicitly, so the known branch equations are no longer merely inventoried but are written as projected shadows of the lifted evolution. Shadow Calculus remains downstream and exact in its proper role as the completion/residual branch.

An immediate engineering application of the same retained-first doctrine is lifted-state symbolic regression. In that application the search does not proceed by genetic-programming word mutation or by an external scalar fitness loop. Projection remains closed during descent, the selector-closed macro-calculus advances the retained state, metriplectic relaxation drives self-termination inside the lift, and the terminal projector opens exactly once at readout. The resulting solver surface already recovers benchmark targets such as x^2 and $\sin x$ from sampled data, showing that Phase Calculus is

not only a theorem-bearing formalism but also a native search architecture.

In this form the formalisation is one manuscript and one hierarchy: Phase Calculus names the whole system, the lifted-state operator calculus is its core, the selector-closed executable macro-calculus is regime-universal across all admissible (W, Δ) on the balanced-window kernel family, universal exact return is forced by non-discharge with the balanced-window family as explicit witness, the mock-theta mechanism is algebraically universal on the balanced corridor with the canonical launch family as verified instance, every physics branch descends by quotient on its declared regime, and Shadow Calculus is its downstream supplement.